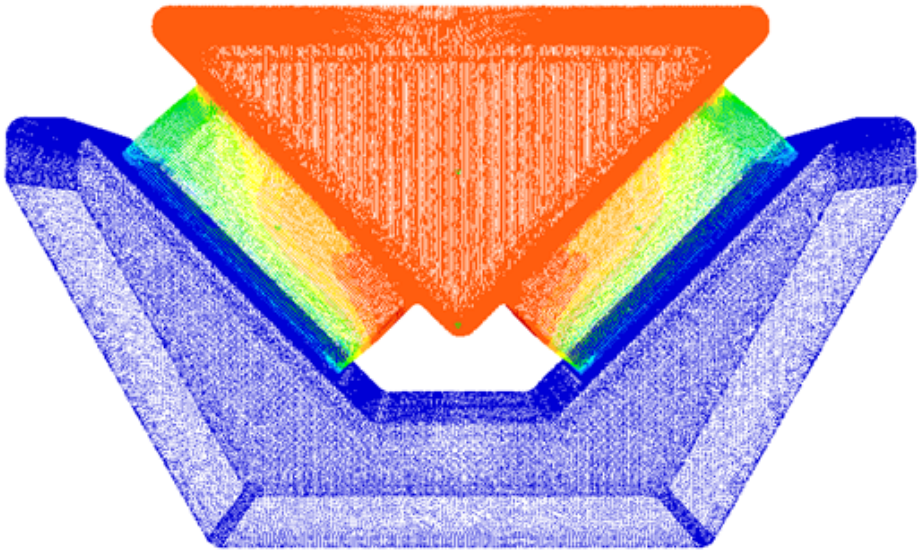
KTH Royal Institute
of Technology

Benchmarking for rubber mount design

Michaela Jankulicova



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Approved	Examiner	Supervisor
2025-05-23	Ellen Bergseth	Ellen Bergseth
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Abstract

The progress in digital technologies is continuous, and with each advancement, the gap between analytical and numerical solutions continues to narrow.

The current development process for rubber mounts is complex and often relies on either the best practices of external suppliers or incremental improvements between design and simulation engineers. This process can be both time-consuming and costly. A benchmarking evaluation was carried out to determine whether the simulation tools currently available to design engineers are sufficiently reliable for early-stage concept evaluation. Two tools—SimSolid and CATIA GAS—were compared to MSC Marc, a simulation tool specifically designed for non-linear materials such as rubber and commonly used by simulation engineers and some suppliers.

Both SimSolid and CATIA GAS utilize linear material models, which were verified for all tested geometries for small displacements . However, this linearity is limited by the shape factor of the rubber material. It must be taken into account, as stiffness changes in linear models due to the bulk modulus become significant when the Poisson's ratio approaches 0.5.

Keywords: Computer modelling, benchmarking, stiffness, non-linear materials, rubber mounts

KTH Industriell teknik
och management

Examensarbete TRITA-ITM-EX 2025:108

Benchmarking för design av gummibussningar

Michaela Jankulicova

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Sammanfattning

Utvecklingen inom digitala teknologier pågår ständigt, och med varje framsteg minskar klyftan mellan analytiska och numeriska lösningar.

Dagens utvecklingsprocess för gummifästen är komplex och baseras ofta antingen på externa leverantörers beprövade metoder eller på stegvisa förbättringar mellan konstruktörer och simuleringsingenjörer. Denna process kan vara både tidskrävande och kostsam. En utvärdering har därför genomförts för att undersöka om de simuleringsverktyg som idag finns tillgängliga för konstruktörer är tillräckligt tillförlitliga för att kunna användas i ett tidigt skede av utvecklingen. Två verktyg – SimSolid och CATIA GAS – jämfördes med MSC Marc, ett simuleringsverktyg som är särskilt anpassat för icke-linjära material som gummi och som används av simuleringsingenjörer samt vissa leverantörer.

Både SimSolid och CATIA GAS använder linjära materialmodeller vilka verifierades för alla modeller för små förflyttningar. Denna linjäritet är dock begränsad av gummits formfaktor och måste alltid beaktas, eftersom styvheten i linjära modeller påverkas av bulkmodulen när Poissons tal närmar sig 0.5.

Nyckelord: Datorbaserad modellering, benchmarking, styvhet, icke-linjära material, gummibussning

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Mami, Jani a Tato, pozrite, uz budem mat vzdelanie.

Michaela Jankulicova
Stockholm, May 2025

Nomenclature

Notations

Symbol	*Description*
A_B	Bulge area (mm^2)
A_L	Load area (mm^2)
d	Displacement (mm)
E	Young's modulus of elasticity (MPa)
E_c	Compression modulus (MPa)
ε	Engineering strain (-)
F	Force (N)
G	Shear modulus (MPa)
K	Bulk modulus (MPa)
K_c	Compression spring rate (N/mm)
K_s	Shear spring rate (N/mm)
ν	Poisson's ratio (-)
ϕ	Material compressibility coefficient (-)
S	Shape factor (-)
σ	Stress (MPa)
t	Rubber thickness (mm)

Abbreviations

CAD	Computer Aided Design
CATIA GAS	Computer-Aided Three-dimensional Interactive Application Generative Assembly Structural Analysis
FEA	Finite Element Analysis
FEM	Finite Element Method

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1. Introduction

This chapter describes the background, purpose, delimitations and framework of the completed thesis.

1.1 Background

Scania CV AB, further referred to as Scania, is a world leading provider of transport solutions [1]. Their main offering includes trucks and buses for heavy transport applications, as well as a wide range of services related to these products [2].

Scania's purpose is aimed to drive the shift towards a more sustainable transport systems. This reflects in their slogan of "Driving the shift" and the four areas where the company is trying to implement their vision [2] are as follows:

1. "Developing and offering premium products, services and solutions for sustainable transport
2. Partnering up to create the enabling conditions for sustainable transport
3. Exploring and accelerating the development of tomorrow's transport systems
4. Managing their impacts on people and planet"

These areas lead to the development of new electric vehicles where new challenges are to be taken into consideration. The Powertrain and Battery Instalation department, at Scania Södertälje works with new designs and usages of rubber mounts, among other things. The term rubber mount will encompass variety of use cases for rubber geometries within this thesis.
Under the current model of operations, the rubber mounts are usually co-developed together with the external suppliers. This means that they design and simulate the component according to Scanias requirements. However, this model of operations can lead to longer lead times due to the ongoing communication between the suppliers and project leaders.

1.2 Purpose

The purpose of this thesis is the visualisation and quantification of CATIA GASs and SimSolids ability to simulate rubber mount properties for different types of rubber geometries. The program (among others) that is used by the simulation engineers and the external suppliers, MSC Marc, is not used by the design engineers because of licensing, ease of use and also the computational time. The following questions were asked:

1. Are these tools, CATIA GAS and SimSolid, capable of estimating stiffness at different displacements/loads for different types of rubber bodies?
   (a) Are there certain specifications the rubber geometries need to fulfil to be able to accurately estimate their stiffness?
   (b) How much does the magnitude of displacement influences the accuracy of the results?
2. Can these tools be used to develop robust and realistic concepts early in the design process?

1.3 Delimitations

The delimitations of this thesis are as follows:

- The amount of geometries analysed was set to 5.
- No considerations for temperature changes of the material from its environment.
- No consideration for the Mullins effect.
- No consideration for error calculation.
- No strain analysis was done during the project.

1.4 Method

Modelling, simulating and evaluating are the three pillars of this project. The tools used for this will be mainly CATIA GAS and SimSolid. This process will include pre-processing of the 3D models, simulating them in respective tool, and post-processing with the results analysis and final evaluation. The verification will be done against results from MSC Marc.

2. Frame of Reference

This chapter presents the theoretical frame of reference that serves as the cornerstone of the executed project.

2.1 Mechanical Properties of Rubber

Polymer materials can be classified into different categories based on the type of links between and within the carbon chains. The definition of rubber is that it is an elastomer with cross-links between the carbon chains that gives the rubber its elasticity and strength [3].

2.1.1 Hyper-elasticity and its non-linearity

The elasticity and strength [3] lead to a different behaviour once force is applied to the material compared to a linear elastic material, such as metal. This behaviour shows non-linearity after certain force is applied as seen in Figure 1. The linear material, also called linear elastic, shows that the stress and the strain are proportional to each other but also that the material comes back to its original form once the force is no longer applied. This is also referred to as loading and unloading paths.

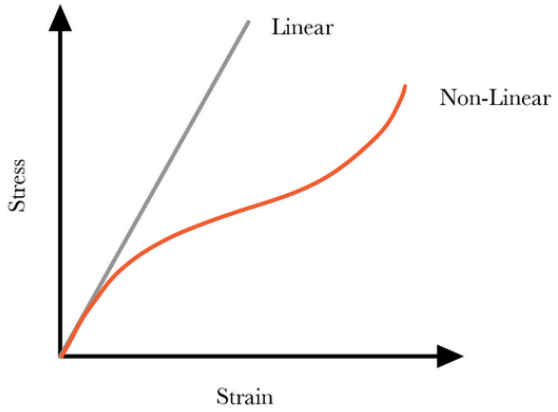


Figure 1: The visualisation of the linear (e.g., steel) and the non-linear (e.g., rubber) behaviour in a stress-strain diagram.

The linearity refers to the proportionality, the elasticity refers to the same loading and unloading path. Thus compared to the hyper-elasticity of rubber, the loading and unloading paths are different, and also dependent on the force applied and the displacement as seen in Figure 2. This adds complexity to analysing hyper-elastic materials since their behaviour is strongly dependent on these two factors.

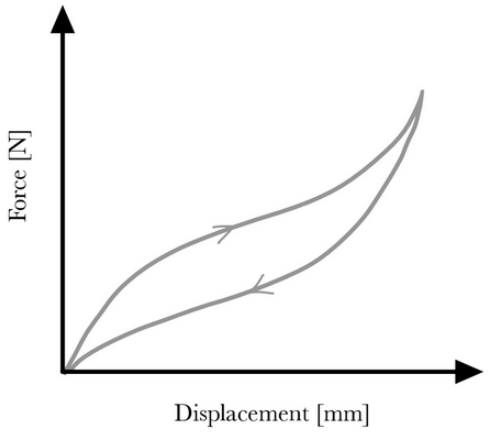


Figure 2: The hysteresis and its change based on the magnitude of displacement and force applied. Loading and unloading gives different paths on the graph.

The difficulty with hyper-elastic materials is the subsequent modelling. The stress-strain relationship of hyper-elastic materials derives from a strain energy density function [4]. There are many hyper-elastic models used in FEA software today, but because of the complexity of these different models, the focus in this thesis was put strictly on the very surface level understanding of the Mooney-Rivlin model.

$$W(I_1, I_2) = C_{10}(I_1 - 3) + C_{01}(I_2 - 3) \tag{2.1}$$

In the Mooney-Rivlin model, equation 2.1, C_{10} and C_{01} are material property values, and I_1 as well as I_2, are invariants but the explanation of those goes beyond the scope of this report.

$$G = 2(C_{10} + C_{01}) \tag{2.2}$$

From the Mooney-Rivlin model, as seen in equation 2.1, it is possible to calculate the Shear modulus of the material with the equation 2.2 by using the material property values.

$$E = \frac{9KG}{(3K + G)} \tag{2.3}$$

The Young's modulus, refers to the ratio between stress and strain on the linear elastic part of the deformation in the axial directions curve. It can be calculated as seen in equation 2.3, where G is the shear modulus and K is the bulk modulus of the material.

Poisson's ratio is a measure for how much the given material is going to change shapes based on deformation direction. A change in the axial and transverse strain after deformation.

$$\nu = \frac{(3K - 2G)}{[2(3K + G)]} \tag{2.4}$$

Shear modulus, compared to Young's modulus, is defined in the same way but instead of deformation in the axial direction, the material deforms in shear.

$$G = \frac{E}{[2(1+\nu)]} \tag{2.5}$$

Bulk modulus is the measure of how resistant a material is to compression.

$$K = \frac{E}{[3(1-2\nu)]} \tag{2.6}$$

$$K_s = \frac{F_s}{d_s} = (\frac{A_L G}{t}) \tag{2.7}$$

Shear spring rate is defined by equation 2.7 [3, p. 262]. Where F_s refers to the force applied in the shear direction and d_s is the shear displacement. A_L is the load area of the rubber and t is the rubber thickness.

The linearisation of rubber properties can be defined by equations 2.3, 2.4, 2.5 and 2.6.

However using equations like 2.7 is dependent on the ratio of thickness to length and that it does not exceed approximately 0.25. That is because after the ratio is too large, the shear deformation due to bending will not be negligible anymore [3, p. 262].

The shape factor is a calculated value based on the load area (A_L) and also restriction of bulging (A_B) of the rubber geometry. This ratio shows the compression modulus of rubber. The smaller the ratio, the more shear modulus dependent the compression modulus is. However, the higher the ratio, the more bulk modulus matters and therefore by extension the materials incompressibility.

$$S = \frac{A_L}{A_B} \tag{2.8}$$

$$E_c = E(1+\Phi S^2) \tag{2.9}$$

The effective compression modulus for a flat sandwich block for a bidirectional strain is defined by equation 2.9 [3, p. 264]. Where Φ is the elastomer compression coefficient and S refers to the shape factor, equation 2.8.

$$K_c = \frac{F_c}{d_c} = (\frac{A_L E_c}{t}) \tag{2.10}$$

The compression spring rate is defined by equation 2.10 [3, p. 262]. Where F_c refers to the force applied in the compression direction and d_c is the compression displacement. E_c is the effective compression modulus.

2.2 Finite Element Method and Analysis

The finite element method and analysis are extensively used in variety of industries.

In the case of more complex problems, it is not always feasible or preferable to solve them analytically. The common practice is to use numerical methods and get a very precise approximation of the result instead. The FEM divides the geometry into infinitely small elements [5],

nodes, that connect to each other. The distance between them is the deciding factor for accuracy and also influences the rate of convergence. There can be structured or unstructured mesh as well as fine or coarse mesh. These are dependent on the program used and what the aim of the computation is.

Where FEM is the mathematical model, FEA is the practical implementation of said model. This means that when talking about FEM in different simulation tools, it is in fact FEA that is being used. Even though the terms might be used interchangeably it is important to distinguish between the two.

3. State of the Art

In this chapter the currently used simulation tools at Scania and their solving methods are presented.

3.1 Design and Simulation Tools

The usage of CAD and simulation tools is a must in development of new products across all industries. The main tools that this project focused on were CATIA GAS, SimSolid and Marc.

CATIA GAS

CATIA GAS (generative assembly structural analysis), further referred to as GAS, is a module available within CATIA 5V program that is used for 3D modelling. This module is specifically used for simulating different kinds of deformations by using an FEA. GAS uses classic FEA that automatically generates mesh that can be either linear or parabolic and its size as well as sag can be chosen by the user for more precise results. The mesh size in connection to the complexity of the geometry influences the speed of convergence during the computation.

The version of CATIA GAS used in this project was version 5-6 Release 2022.

SimSolid

Simsolid is structural analysis simulation tool that is using engine that is based on breakthrough extensions to the theory of external approximations. It uses fully featured and not simplified solid geometry models. However, it does not use mesh like a classical FEA 6. Simsolid is very user friendly and includes many automatic functions for streamlined workflow.

The version of SimSolid used in this project was 2023.0.0.211.

MSC Marc

MSC Marc, further referred to as Marc, is a program specifically used for non-linear simulations by the simulation engineers at Scania.

Mesh creation is done in another program called HyperMesh that is beyond scope of this report. This mesh is then imported to Marc where it gets applied to the model. As already mentioned, Marc is made for use with non-linear materials and has a wide variety of simulation options.

For this project, the Mooney Rivlin model was chosen. It is very clear what models are used at every point of the process and how the results are calculated. However Marc is also much more complex to use and not nearly as user friendly as SimSolid, as well as there are limited amount of licences/access to Marc for the design engineers. Marc is also the program with which the verification is going to be done.

The version of MSC Marc used in this project was 2024.1.

4. Methodology

This chapter describes step by step the different methods used for fulfilling the projects aim.

4.1 Geometry selection and design

All three geometries (A, B and C), were designed pre-project with the aim to best suit the projects needs. The design tool used was CATIA 5V. Certain models were adjusted throughout the project once the need presented itself.

4.1.1 Geometry A

Geometry A was chosen for its simplicity that it brings to each step of the simulation process. The subsequent number after the letter A refers to the height of the rubber in millimetres. This geometry has three components in the assembly. Top plate, bottom plate and the rubber block in the middle, as seen assembled in Figure 1, 2a and 2b.

Each height was chosen with careful consideration to the shape factor. The aim was to be able to compare how much the shape factor influences the accuracy of the results with the exact same initial conditions. The exact measurement of each rubber elements can be found in Table 1.

Table 1: Geometry A measurements

Geometry name	Height [mm]	Length [mm]	Width [mm]
A.20	20	50	50
A.05	5	50	50
A.01	1	50	50

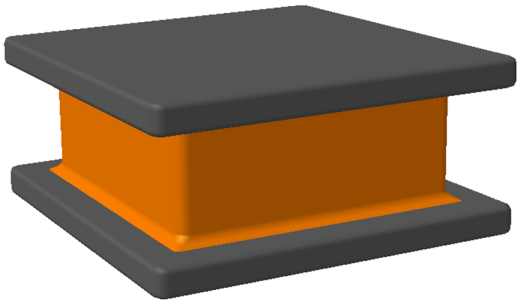
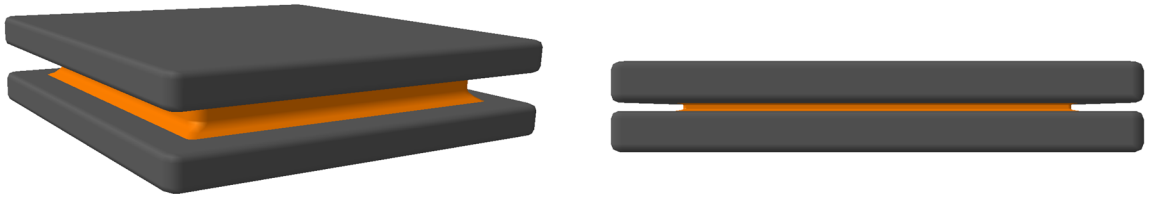


Figure 1: Geometry A.20



(a) Geometry A.05 (b) Geometry A.01

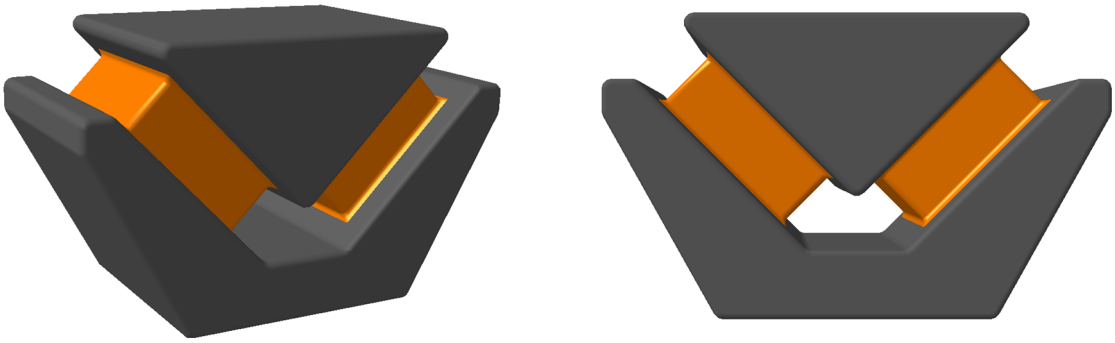
Figure 2: Geometry A

4.1.2 Geometry B

This geometry was chosen to reflect a real life example of a rubber mount design. It consists of four different elements that are assembled (Figures 3a, 3b). These elements are two rubber blocks, one triangular geometry and one base. The angle between the bottom of the base and the rubber blocks is 45 degrees. Their sizes can be found in Table 2.

Table 2: Geometry B measurements, both sides are the same

Geometry name	Height [mm]	Length [mm]	Width [mm]
B	20	50	50



(a) Geometry B: side view (b) Geometry B: front view

Figure 3: Geometry B

4.1.3 Geometry C

The flanged cylinder bushing was chosen because it has two different loading parts both in the radial and the axial directions. This geometry is simplified real life example just as Geometry B. This geometry consists of six elements. Top plate, sleeve, top and bottom rubber bushings, bracket and a bottom plate. The assembly can be seen in Figure 4.

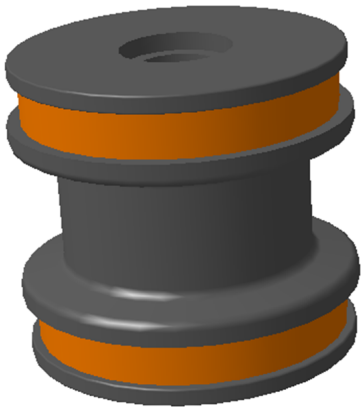


Figure 4: Geometry C

The measurements of the rubber bushings can be seen in Figure 5 where half of the model is visualized.

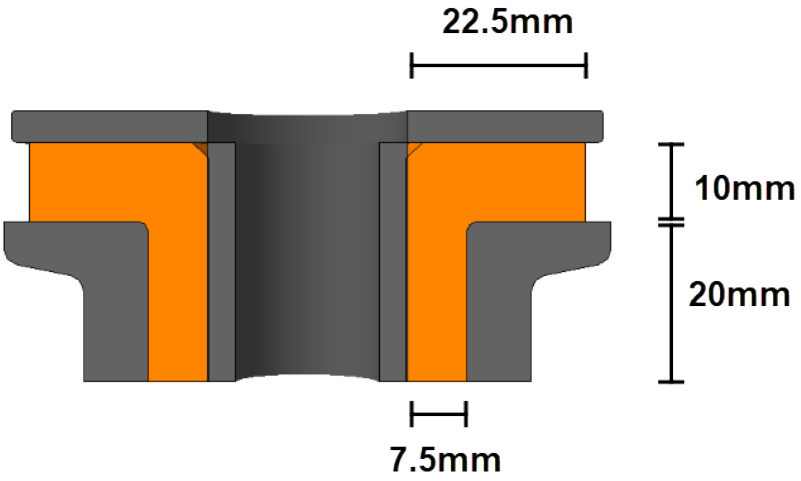


Figure 5: Measurements of a half-model of Geometry C that is symmetrical at the bottom.

4.2 Manual calculations

The analytical simplified results were manually calculated to create a baseline for the simulated, numerically calculated results. This, however, is only valid for Geometries A and B since the rubber shape is very simple. For Geometry C there is much more nuance to its shape.

There are several factors that need to be considered when calculating stiffness of rubber by using analytical methods. As mentioned in 2, rubber is a hyper elastic material which means there is not only one value for the same displacement value because of different loading and

unloading paths.

Based on the deformation type, displacement in z- (compression) or y-directions (shear), different material properties need to be linearized and calculated.
Firstly, the shape factor had to be calculated. The simplified version is defined as seen in equation 2.8.
Compression spring rate is defined in equation 2.10 as well as shear spring rate in equation 2.7.
The values calculated from these equations can be seen in Table 3.

Table 3: Geometry A: manual calculations

Geometry name	$A_L[mm^2]$	$A_B[mm^2]$	$S[-]$	$K[MPa]$	$G[MPa]$
A.20	2500	4000	0.63	5919.68	1.18
A.05	2500	1000	2.50	5919.68	1.18
A.01	2500	200	12.50	5919.68	1.18

Table value for ϕ can be found in [3, p. 265,Table 8.1]. Further, the effective compression modulus E_c was calculated according to equation 2.9 and by using the equation 2.10, we get the value for the compression spring rate K_c, or better said, stiffness of the rubber during compression. All of these values are summarized in Table 4.

Table 4: Geometry A: Stiffness calculation

Geometry name	$E_c[MPa]$	ϕ	$K_c[N/mm]$
A.20	4.3	0.54	537.626
A.05	15.54	0.54	7769.563
A.01	303.235	0.54	758087.313

4.3 Pre-Processing of the Models

Pre-processing needs to be done to successfully simulate the deformations and thereafter extract the results. It contains the following steps.

Assuming there is already pre-existing 3D model of the geometry, depending on which program it was designed in, the specific file type will have to be downloaded. Because GAS is a module within CATIA, the program that all the models were designed in, the simulation file creation is the simplest and very straightforward. This is done directly from the file within CATIA. SimSolid and Marc both need .stp file that can be downloaded from CATIA. But are not limited to this file type in general.

After uploading the model to one of the programs, the next step of the process can begin.

4.3.1 Mesh generation and contact conditions

Mesh is a method within FEM and is used in FEA programs. The mesh is automatically created in GAS and chooses an arbitrary size of the grid based on the different parts of the model.
Contact conditions is used as an umbrella term for both the analysis connection but also the connection property for those.

There needs to be initial relationship decided before its more specific expected behaviour is decided. Those differ between SimSolid and GAS from how they are created to how they are changed and processed.

GAS

In GAS the first step for connection creation is to choose the analysis connection. In this project only the general analysis connection was used since it uses the faces of the model. Their purpose is to connect any part of an assembly with or without a handler point on an assembly model.

Once all of these general analysis connections are established, the property types can be chosen. By choosing a property type, the wished behaviour of the connection is chosen. This can vary greatly and therefore all of the main options used are presented in a table 5.

Table 5: Connection properties GAS

Connection property	Description	Icon
Rigid	Link between two bodies which are stiffened and fastened at their common boundary. This makes them behave as infinitely rigid.	
Slider	Link between two bodies which are constrained to only move in the local normal direction at their common boundary. This allows them to slide relative each other in the local tangential plane.	
Contact	Link between two bodies which are prevented from interpenetrating at their common boundary. They are allowed to move arbitrarily relative to each other as long as they do not come into contact within a user-specified normal clearance. A friction ratio can be defined as well as no sliding condition can be set.	
Fastened	Link between two bodies which are fastened at their common boundary and will behave as if they were a single body. The difference to the rigid connection is that here the corresponding nodes of two compatible meshes are merged together.	
Pressure fitting	Uses assembly surface contact constrained as a support. This connection is the link between two bodies which are assembled in a Pressure Fitting configuration, when there are interfaces that overlap between both parts. Along the surface norma, the connection behaves as a cntact condition with negative clearance value (positive overlap). This value lies in the tangential directions where both parts are linked together.	
Smooth	Link between two bodies which are fastened together at their common boundary. This allows them to behave as if their interface was soft.	

SimSolid

SimSolid has an automatic function that suggests the connection tolerances with the gap size and penetration size which can be also user adjusted. The connection resolution level can also be chosen between normal, increased and high. The recommended level is normal unless the

contact area is small in relation to the overall size of the connecting parts [7].

The connections within the imported model are created with an automatic function or by manually choosing. If the automatic function is chosen, the user can adjust if they are correct for the aimed simulation. This thesis focused on regular connections and will not go into detail about the other connection possibilities.

There are several types of regular connections with different properties and their descriptions are in Table 6.

Table 6: Regular connections SimSolid

Connection property	Description	Color
Bonded	The loads between parts are fully transferred and the bonded components never separate.	
Sliding without friction	Parts can slide relative to each other in the local tangential plane, wile not allowing inter-penetration or seperation of the connected surfaces. Only normal forces to the surface are transferred between the parts. Assumes frictionless sliding surface.	
Separating	The parts may partially or completely separate from each under under the load. The forces transfer through the connected regions. Friction coefficient is chosen and defines the sticking/sliding threshold. This is non-linear solution and adds computational time compared to the other connections.	

4.3.2 Material properties

Material properties input into the program differ based on the model used for the calculations. SimSolid and GAS use linear models which among other values, need Young's modulus and Poisson's ratio for the materials used. Because verification is done against Marc, the Mooney-Rivlin model needs to be translated into the linear one.

The linearization follows the steps described already in chapter 2. The material property values C_{10} and C_{01} were acquired directly from the data input into Marc by the simulation engineers. The shear modulus was thereafter calculated by using equation 2.2. The bulk modulus was calculated within Marc by its automatic function. The Poisson's ratio was calculated by using equation 2.4. Lastly, Young's modulus was calculated by using the equation 2.3. Their values can be found in table 7.

Table 7: Material properties

$C_{10}[MPa]$	$C_{01}[MPa]$	$G[MPa]$	$K[MPa]$	ν	$E[MPa]$
0.474	0.118	1.184	5920	0.4999	3.5518

4.3.3 Boundary Conditions

Which boundary conditions need to be set varies based on many factors but one of the most important ones is the form of deformation. This thesis focused on compression in the normal direction to the surface and shear deformation.
The boundary conditions used in the different simulation cases can be found in Table 8 for GAS and in 9 for SimSolid.

Table 8: Boundary conditions GAS

Name	Description	Icon
Clamp	All points on the applied surface or curve are blocked in the subsequent analysis.	
Surface slider	Allows points of a surface to slide along a coinciding rigid surface.	
User-defined restraint	This restraint allows the user to fix any combination of available nodal degrees of freedom on arbitrary geometries.	

Table 9: Boundary conditions SimSolid

Name	Description	Icon
Immovable support	Enforce zero translational displacements in all directions on faces, edges, vertexes, and/or spots.	
Slider support	Enforce zero displacement in normal to the surface direction on faces and areal spots.	

The boundary conditions defined by user can also be seen in Table 10 as a change of displacement in a specific global direction.

Table 10: Boundary conditions for all three cases

Displacement direction	Compression	Shear	Extension
x	0	0	0
y	0	Can be either positive or negative	0
z	Negative value	0	Positive value

Compression

Special boundary conditions for compression was zero displacement in y- and x-directions whilst the aimed displacement was in negative z-direction.

Shear

In the simulation of shear displacement, there is a chance for bending. The boundary conditions therefore need to be zero displacement in z- and x-direction.

Extension

Special boundary conditions for compression was zero displacement in y- and x-directions whilst the aimed displacement was in positive z-direction.

4.4 Simulation

Each geometry and each scenario had given displacement increments that were simulated.

Geometry A

Geometry A was simulated in two different scenarios with boundary condition clamp on the bottom of the bottom plate and user-defined restrained for all directions on the top of the top plate. The contact properties were set to rigid between the rubber body and both the top and bottom plates. The connection property placement is visualized in Figure 6 and the specific connections chosen can be seen in Table 11. Two different scenarios refer to both compression and shear. In Marc the simulation results are for fully bonded connections.



Figure 6: The faces that can have a specific connection type chosen for geometry A.

Table 11: Geometry A: Simulations

Colour	Connection property (GAS)	Regular connection (SimSolid)
	Rigid	Bonded
	Rigid	Bonded

Geometry B

Geometry B includes boundary conditions on the bottom but also the sides to make them infinitely rigid and avoid moment in the corners. Those were set to clamped and user-defined retrain for all directions was set on the top of the triangle. The contact properties were set to rigid between the rubber blocks and the base and fastened connection between the rubber blocks and the triangle. The possible connection faces can be seen in figure 7 and the connections summarized with their respective colors can be seen in Table 12. In Marc the simulation results are for fully bonded connections.

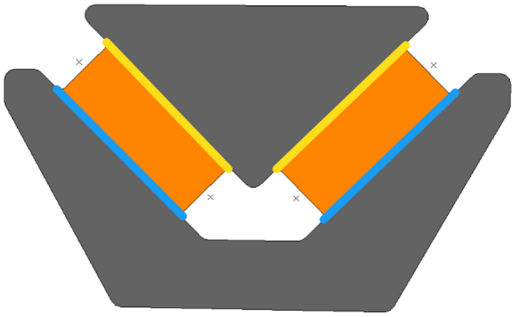


Figure 7: All the possible connections for Geometry B.

Table 12: Geometry B: Simulations

Colour	Connection property (GAS)	Regular connection (SimSolid)
	Fastened	Bonded
	Rigid	Bonded

Geometry C

Geometry C was more complex based on the shape and amount of connections as well as the boundary condition settings. The simulations themselves and their connections can be seen in Table 13.

All the possible connection faces can be seen in figure 8. Initially there are two different simulations that we will call Simulation 1 and Simulation 2. Simulation 1 refers to a fully rigid geometry where the rubber is vulcanized to the metal and Simulation 2 refers to a simulation where a certain amount of surfaces have possibility of friction. When talking about friction coefficient, the value of 0.4 was used as a standard based on the knowledge of the simulation team. The connections used in each simulation are noted in Table 13.

For Simulation 1 the rubber is also fully bonded in Marc however for Simulation 2 the rubber is loose with friction.

Because of the higher complexity of this geometry based on the amount of connections and the shape, certain simulations took longer than 6 hours of computing time which for the aim of this project is unreasonable. Therefore the geometry was simplified to examine radial and axial cases separately. In total there are four different cases. Two in radial direction and two in axial. Each connection set up can be found in Table 14.

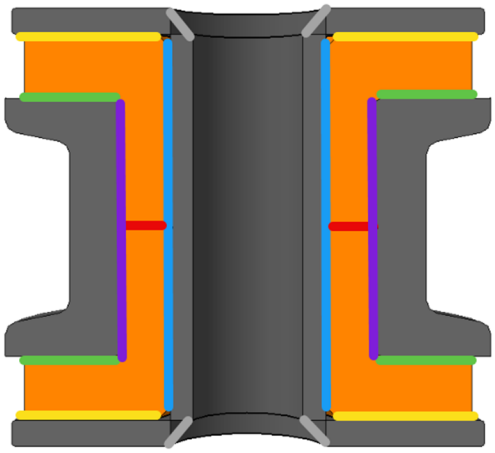
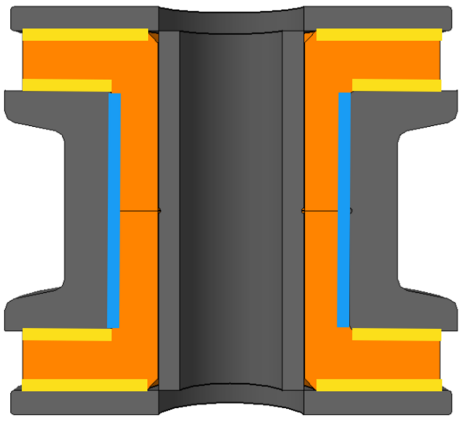


Figure 8: All the possible connections for Geometry C.

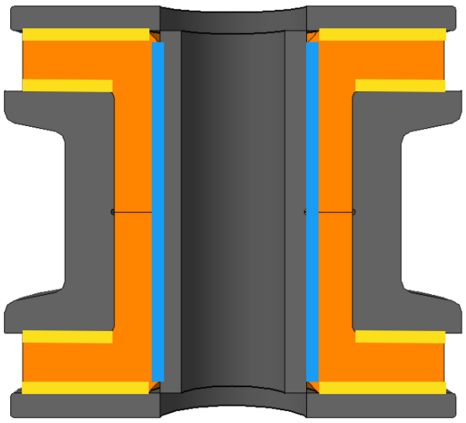
Case 1 and Case 2 are displacements in the axial direction where the aim is to see the stiffness in the flange part of the geometry. Case 3 and Case 4 are displacements in the radial direction where the aim is to see the stiffness in the cylindrical part of the geometry. The connections for all four cases can be seen in Figure 9 and the explanation of the type of connection can be found in Table 14.

Table 13: Geometry C: Simulations

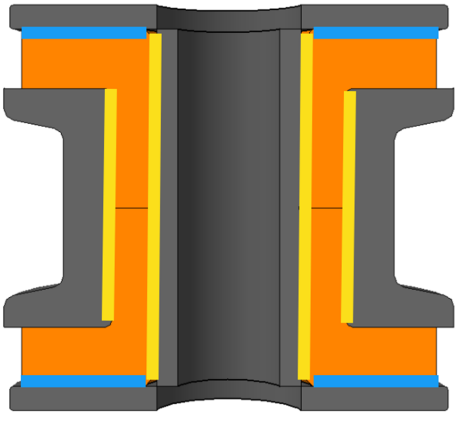
Color	Simulation 1	Simulation 2
	Rigid/Bonded	Rigid/Bonded
	Rigid/Bonded	Contact condition/Separating
	-	-
	Rigid/Bonded	Contact condition/Separating
	Rigid/Bonded	Rigid/Bonded
	Rigid/Bonded	Contact condition/Separating



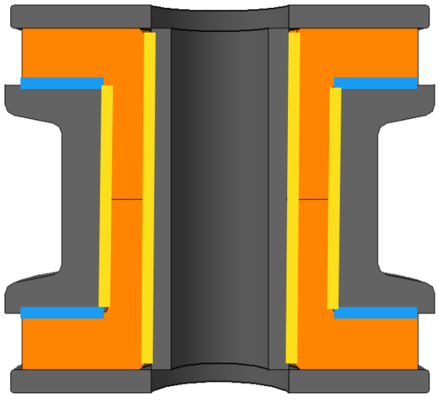
(a) Case 1



(b) Case 2



(c) Case 3



(d) Case 4

Figure 9: The four simplified cases for Geometry C.

Table 14: Geometry C: Simulations

Color	Connection
	Sliding with friction
	Sliding without friction

Simulating the deformations differs based on the simulation tool but the calculation is straightforward in that in both SimSolid and GAS the simulation gets activated and then the results are displayed and then further noted into a separate MS Excel sheet where all the data was saved.

In GAS the reaction sensor gives information on the reaction forces in the chosen connection property as well as moment in all three directions. It is important to note that the right coordinate system needs to be chosen for these values to be valid. In SimSolid the reaction forces are read from reaction/contact force menu where the focused components can be chosen.

Poisson's ratio influence on convergence

When Poisson's ratio is nearing 0.5, problems are caused in computation because of bulk modulus going towards infinity. Because of this, the rubber becomes infinitely stiff. To examine this influence further, several simulations were done on Geometry A.20 with various Poisson's ratio values to see the difference in the results but also compare the differences between GAS and SimSolid. The values simulated were $0.4999, 0.490, 0.470$, all for compression.

4.5 Post-Processing of the Simulations

Each displacement increment for each simulation scenario was recorded into an MS Excel file, it is worth mentioning that this is strictly very small displacements. There is awareness of the simulations not being accurate with bigger displacements with its respective reaction force. These were then imported into a MATLAB script where the results were plotted for analysis and evaluation. At this step of the process the changes in behaviour between different geometries could be also noted and further analysed if some discrepancies were found.

The analysis was done by, as stated above, plotting the force/displacement for the given scenario in comparison to each program. The angle of the curve is the stiffness of the rubber in N/mm and can be directly compared between the three tools. The visualisation of the results themselves is presented in the results and their evaluation is then further discussed in the discussion.

5. Results

This chapter showcases the results acquired during the methodology process of this project.

5.1 Manual calculations

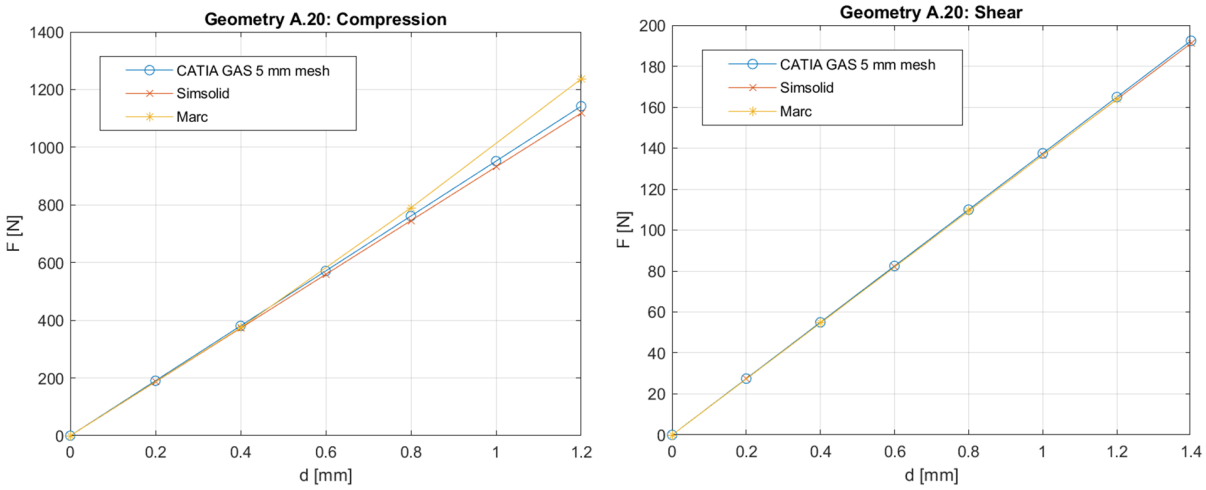
Because of the complexity of rubber and its non-linear behaviour these calculations were not reliable to use as the true analytical value for the same deformations.

5.2 Stiffness

In Figure 1, the plots represent the values acquired during the methodology portion of this project from the simulations as a resulting forces through a given displacement which gives us the stiffness in N/mm.

The stiffness for Geometry A

The results for Geometry A can be seen in Figures 1a, 2a, 3a for displacement in z-direction (compression) and in Figures 1b, 2b, 3b for displacement in y-direction (shear). It can be seen that the plots in Figure 1a show linearity with Marc for up to 0.8 mm in compression where thereafter the non-linearity of Marcs model starts to diverge in the results. The shear displacement in Figure 1b is fully linear.

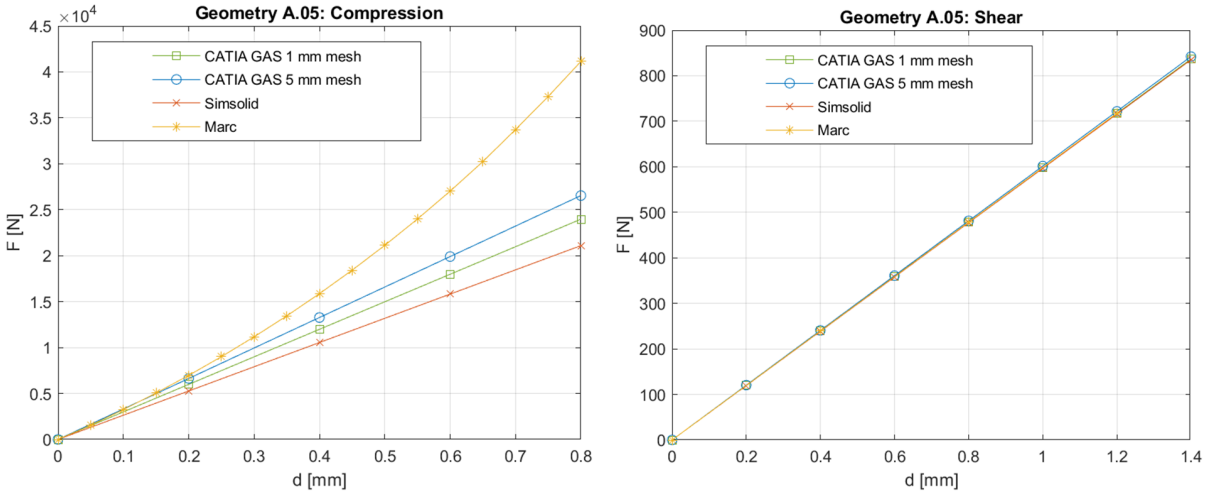


(a) Compression

(b) Shear

Figure 1: Compression of the 20 mm thick rubber block and shear for the same geometry.

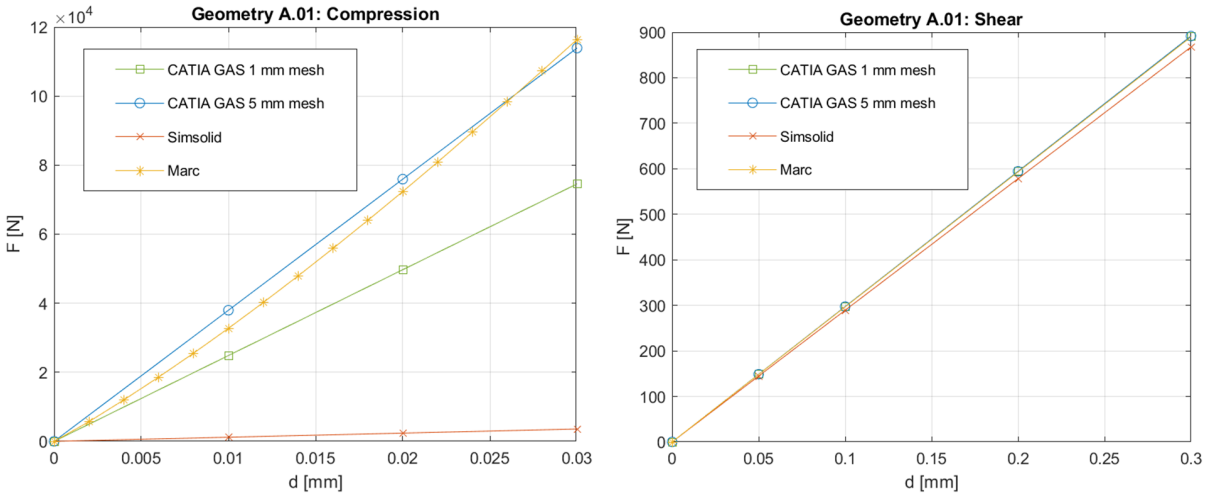
However, in Figure 2a the linear portion becomes smaller and only up to 0.2 mm in compression displacement before Marc results start to diverge. The shear, as seen in Figure 2b, is still linear for all three programs.



(a) Compression (b) Shear

Figure 2: Compression of the 5 mm thick rubber block and shear for the same geometry.

For the most extreme case of shape factor 12.5, there is almost no linearity between Marc and the two other programs. This can be seen in Figure 3a where the 1 mm mesh CATIA GAS and SimSolid results diverge right from the very beginning. The shear however is still linear, as can be seen in Figure 3b.



(a) Compression (b) Shear

Figure 3: Compression of the 1 mm thick rubber block and shear for the same geometry.

The Figure 4 shows how the difference of shape factor drastically influences the stiffness of the given geometry. The stiffest geometry is the one with the highest shape factor.

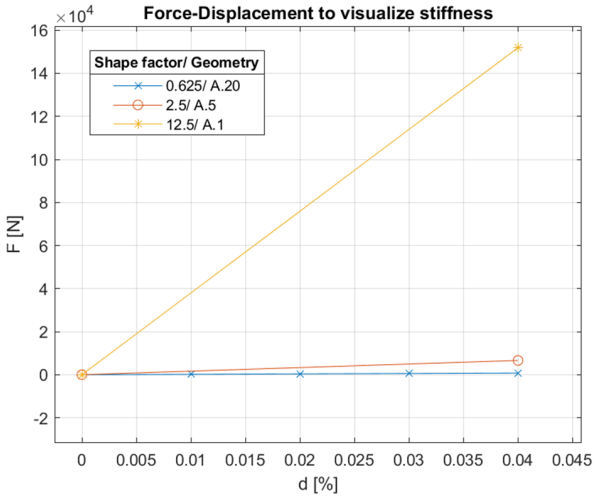


Figure 4: The shape factor as a function of force applied and the percentile displacement for the different thicknesses of the rubber in Geometry A.

The stiffness for Geometry B

The stiffness for Geometry B can be seen in Figure 5 where both compression and shear were present during the displacement. As seen in Figure 5, the plots are linear for up to 1.2 mm before Marcs results begin to diverge.

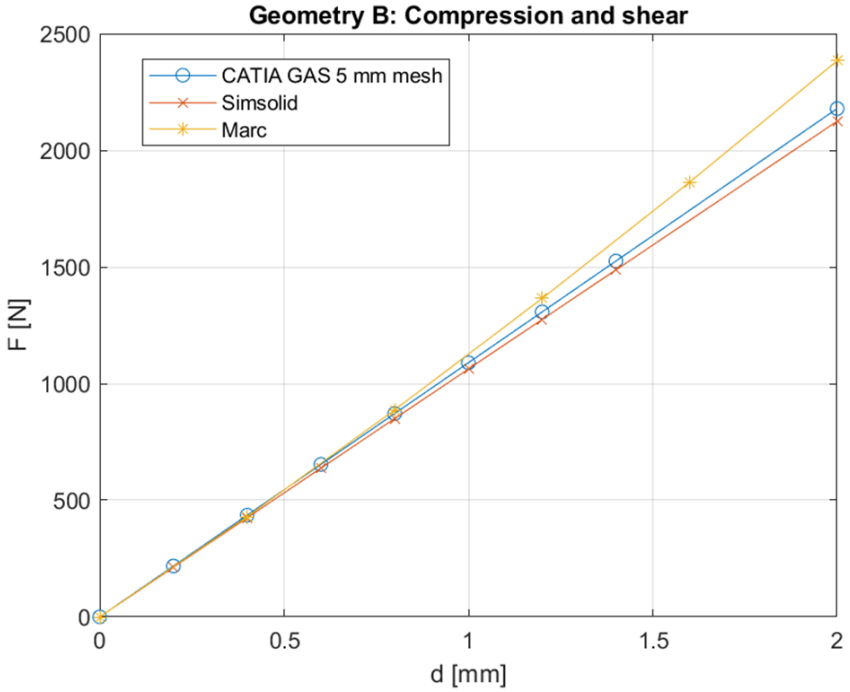
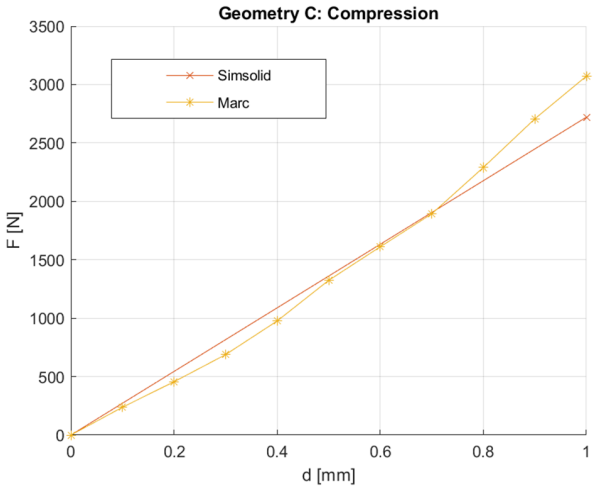


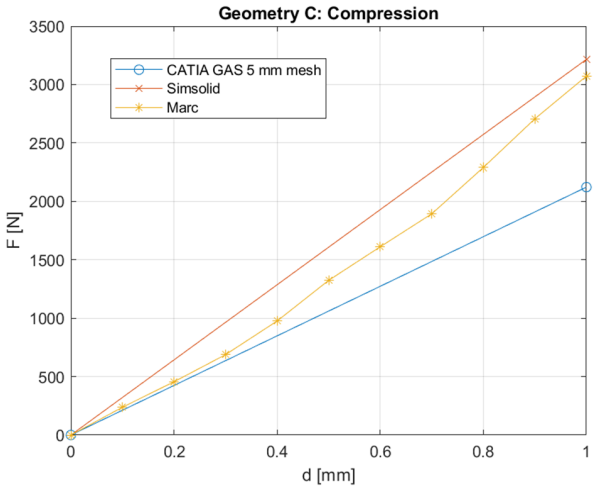
Figure 5: Compression and shear on the two rubber blocks of the geometry.

The stiffness for Geometry C

The results for Geometry C can be seen in Figures 6a,6b, 7a and 7b as well as in Figures 8a and 8b. The graphs not showing GAS as one of the results are ones that either failed or the computational time took longer than 6 hours.



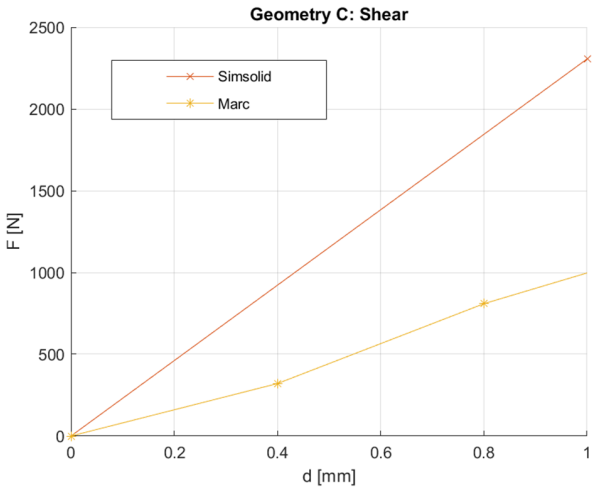
(a) Case 1



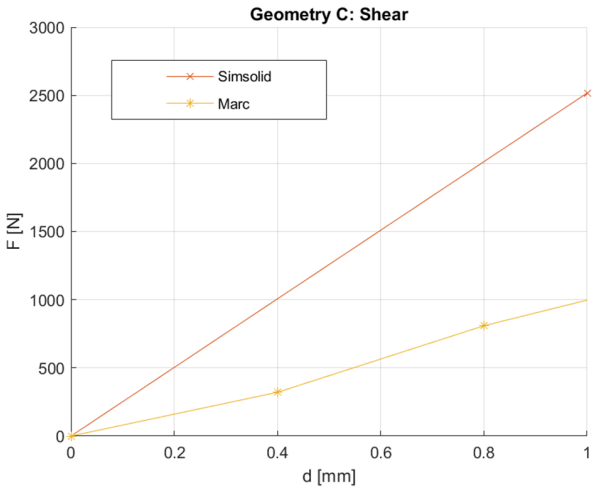
(b) Case 2

Figure 6: The axial displacement for the two cases as a simplification of the earlier simulations.

The Cases 1 and 2 showed linearity for up to 0.1 mm in Figure 6a and up to 0.1 mm in Figure 6b, however only for CATIA GAS becuase SimSolid diverges instantly.



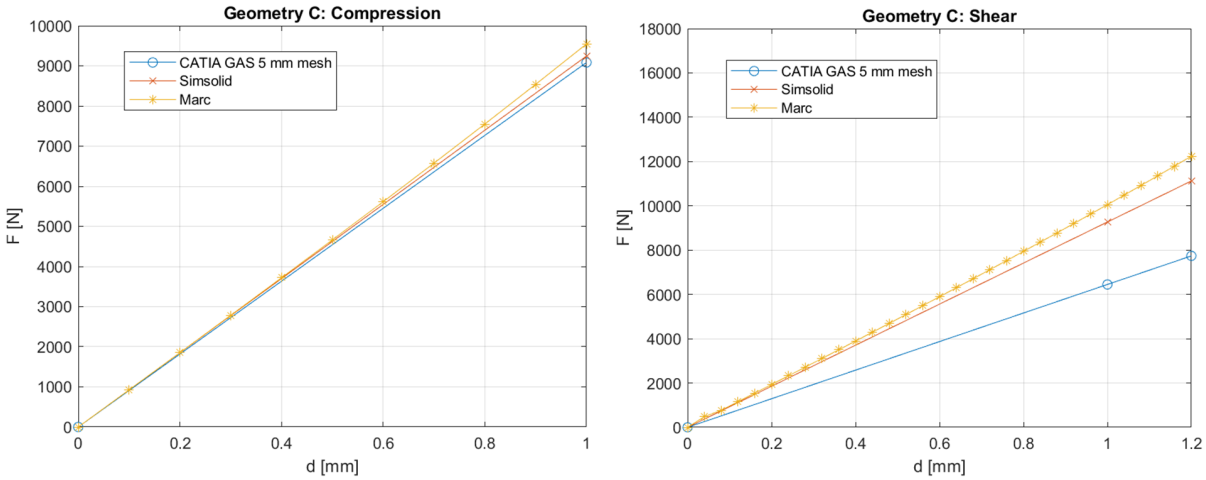
(a) Case 3



(b) Case 4

Figure 7: The radial displacement for the two cases as a simplification of the earlier simulations.

The Cases 3 and 4, as seen in Figures 7a and 7b diverge right away from Marcs results. The fully rigid simulation, as seen in Figure 8a, is linear through the whole simulation, which is up to 1 mm in the axial displacement. In Figure 8b, GAS diverges right away however SimSolid is linear with Marc for up to 0.8 mm in the radial displacement.



(a) Axial displacement.

(b) Radial displacement.

Figure 8: Simulation 1: fully rigid simulations for both axial and radial directions.

In Figure 9, we can see SimSolids results diverge right away but GAS and Marc have common linearity for up to 0.7 mm.

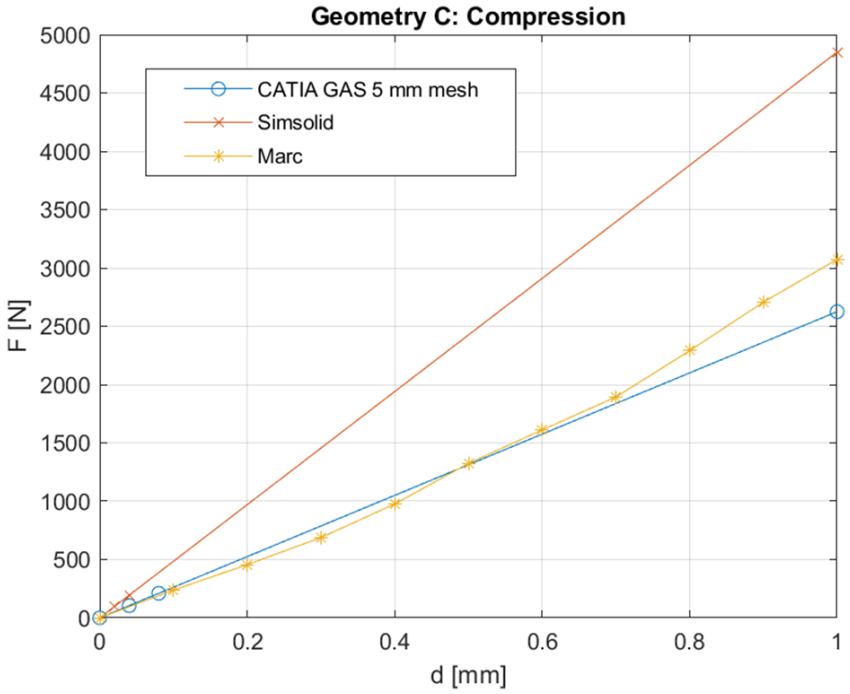


Figure 9: Simulation 2: friction on certain surfaces.

All the simulation settings and also their results for all 5 geometries can be found in Appendix A, B and C.

Influence of Poisson's ratio on stiffness

Change in Poisson's ratios value is shown in Figure 10. We can see the change of behaviour in both GAS and SimSolid is dependent on Poisson's ratio and how much the two programs converge.

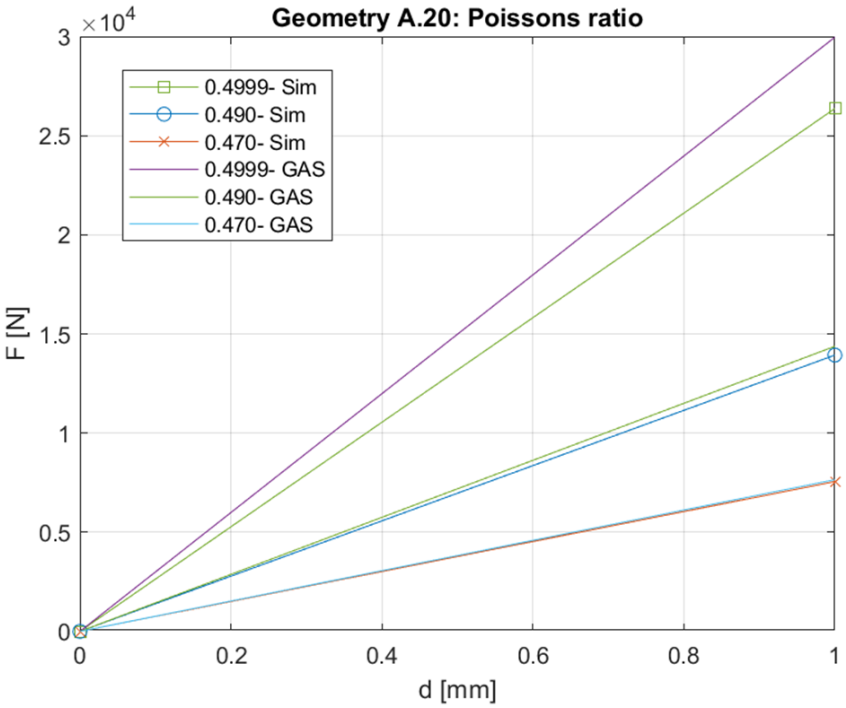


Figure 10: Different Poisson's ratio values and their influence on the results in the two programs.

The simulation setting can be seen in Appendix D.

6. Discussion and Conclusion

This chapter discusses the results acquired and draws conclusions of this thesis.

6.1 Discussion

Because of the complexity of hyper-elastic materials the manual calculations proved unreliable and therefore discussion will be aimed at comparing the two programs to Marc.

Three main factors should be considered when choosing the most optimal work flow for rubber mount design from a design engineers point of view. The use case of the rubber mount and therefore its goal shape, the ease of use of the program and the most important, the factor that was examined in this thesis, the accuracy of the results in comparison to Marc. A trend was visible early on that the shape factor, as stated in chapter 2 2, influences the results accuracy. This can be seen especially when comparing the different sizes of rubber in Geometry A, in Figures 1a, 2a and 3a. The lower the shape factor, the more accurate the results are, Figure 1a, from the linearized model because of the influence of bulk modulus on compressibility of rubber.

As mentioned above, shape factor is the biggest indicator for the result accuracy. Therefore the workflow should begin by doing an analysis of the goal design of the geometry based on the size of the rubber (which is connected to the indented function) and a simplified shape factor calculation or at least a reasonable assessment. This was however only confirmed on the simpler shape of rubber of Geometry A and Geometry B.

The accuracy is very dependent on a correct set up of the model in each of the programs. The boundary conditions as well as the connection types within the assembly are a key to possibly reliable results.

Shape factor for Geometry A.20 showed accuracy up to 1.2 mm of displacement during compression, Figure 1a, which is 6% of the total thickness of the rubber . However, shape factor for Geometry A.05 showed accuracy up to only 0.2 mm, which is 4% of the thickness of the rubber, Figure 2a. As mentioned earlier, the higher the shape factor the more bulk modulus influences the behaviour of the rubber and that could be seen in the results for Geometry A.01 which showed accuracy only up to 0.03 mm for compression, Figure 3a, which is only 3% of the thickness. All of the different thicknesses were however linear during the shear deformation for the small displacements.

The results for Geometry B showed the same accuracy as results for Geometry A for compression and that is up to 6% accurate of the total thickness of one rubber block.

The results for Geometry C are divided into four different cases and two different separate simulations. During the cases certain simulations were not able to be performed in GAS due to failure of the simulation or too long computational time, 6 hours or more.

Simplified cases

Cases 1 showed accurate results up to 0.8mm in SimSolid, Figure 9a, however, GAS never got a result. Case 2 showed very high accuracy in Gas for up to 0.3 mm displacement in the axial direction, Figure 9b.

Case 3 and Case 4 only have results for SimSolid and those completely diverged from Marc, Figures 9c and 9d. This can be connected to the possibly high shape factor for such geometry, even though not analytically calculated, because of the bulging area being very small and the small thickness of the rubber compared to the loading area in the radial direction.

The most interesting results were in simulation 1 for the fully rigid rubber body. For compression the accuracy was high, up to 1 mm displacement for both GAS and SimSolid, Figure 8a, and for shear accurate up to 1.2 mm displacement, but only in SimSolid, Figure 8b. GAS results were not satisfactory.

Simulation 2 with certain friction surfaces in both GAS and SimSolid showed accuracy in GAS for up to 0.8 mm displacement and lacked accuracy in SimSolid , Figure 9.

Poisson's ratio

The shape factor can however be influenced by adjusting the Poisson's ratio. Since Poisson's ratio is between 4.999 and 5 for rubber materials and is both shear and bulk modulus dependent, changing its value will change also how the rubber will behave in a simulation. As seen in equations 2.6 and 2.5, the bulk modulus is much more Poisson's ratio dependent because of its connection to rubbers incompressibility. The influence of Poisson's ratio in both GAS and SimSolid can be seen in Figure 10. The lower the value the more the two programs converge which can indicate a possibility of change of Poisson's ratio in the future for specific results.

6.2 Conclusion

SimSolid is user friendly and has many automatic functions that ease the workflow rapidly. GAS has a specific step-by-step approach where the user has a full control over what is done and how it is done. The use of material properties that were linearized from MSC Marc is needed.

Both SimSolid and CATIA GAS are capable of estimating stiffness at different, small displacements for various types of rubber bodies mostly if fully bonded connections are set. The specifications they need to fulfil are rubber shape dependent, more specifically the bulging and load areas ratio. The bigger the shape factor the less accurate the results will be. This leaves a lot of reasoning up to the design engineers to decide the best approach based on the type of rubber mount they are designing.

The magnitude of the displacement needs to be within very small margin of the total thickness, as seen in for example compression results from Geometry A.20, of the rubber but also heavily dependent on the shape as mentioned above. During simulations of Geometry A.05 and A.01 it was clear how much SimSolid is influenced by the higher shape factor and how inaccurate it becomes the higher the shape factor becomes.

The tools examined in this thesis can indeed be used to develop robust and realistic concepts early on in the design process as long as a proper analysis is done beforehand by the design engineers about the use, shape and size of the rubber elements.

7. Future work

For future work, it would be recommended to examine if GAS and SimSolid are capable of estimating strain in the different geometries as well as if they are capable of calculating accurately the stress distribution in the rubber mount and by extension make it possible to optimize the rubber shape for reduced strain.

Deeper analysis of mesh size in GAS and its influence on the accuracy of the simulations according to correct FEM practices would also be recommended.

Another recommendation would be to examine how does the design of the contact surfaces impact the simulation possibilities with fiction and other connections in GAS.

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Appendix A: Simulation results Geometry A

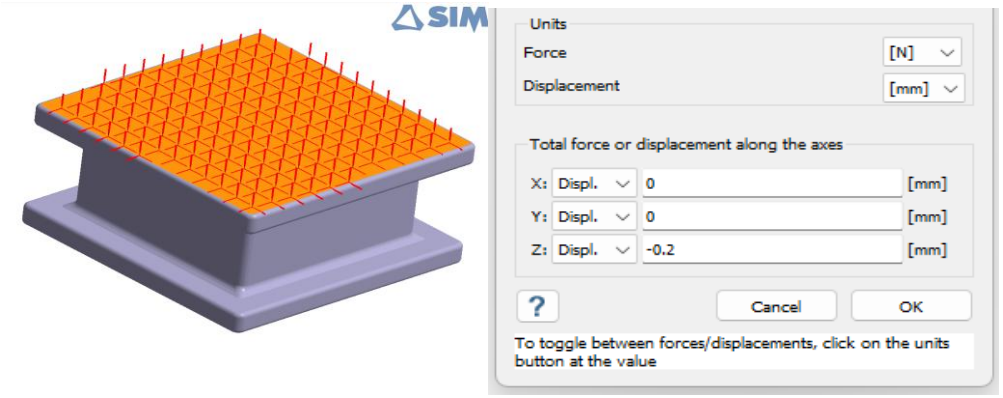
Geometry A- Block 20mm

Simsolid Settings		
		Units
Gap	0.096695	mm
Penetration	0.048348	mm
Connection resolution	Normal	-
Connect unwelded parts only	No	-
Regular connections	Bonded-top and bottom	-
Contact condition	Bonded	-
Refinement level	Steel-standard, Rubber-high	-
Max number of adaptive solutions	3	-
Adapt to thin solids	Yes	-
Adapt to features	No	-
Displacement and conditions	y=x=0 mm and z negative disp	

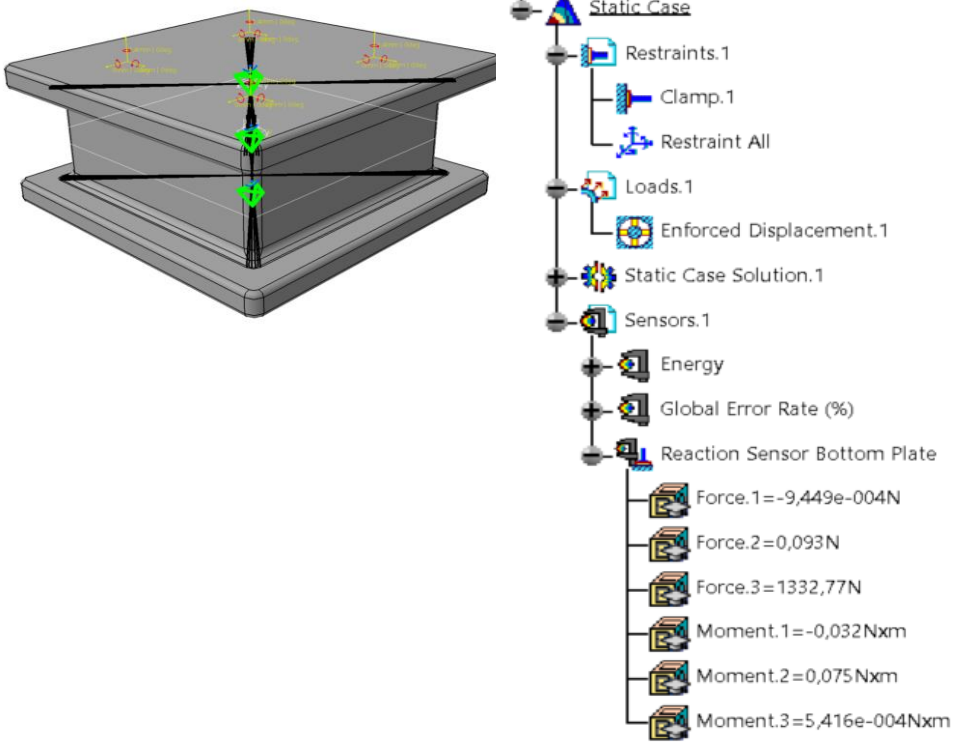
CATIA GAS Settings	
Analysis connection	General analysis connection both top and bottom
Properties	Rigid connection both top and bottom
Restraints	User-defined: All
	Clamp-bottom plates bottom
Loads	Enforced displacement-Translation 3
Sensor	Reaction sensor on the bottom rigid conn.
Force	Force 3

Youngs modulus	4.45	MPa
Poissons ratio	0.495	-

Simsolid-Rubber 60IR	F [N]	0	227.2	454.4	681.60	908.80	1136.0	1363.2	1590.4
	d [mm]	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4



GAS-Rubber Special	F [N]	0	228.729	457.458	686.187	914.916	1143.646	1372.375	1601.104
5mm mesh, 0.5 sag	d [mm]	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4



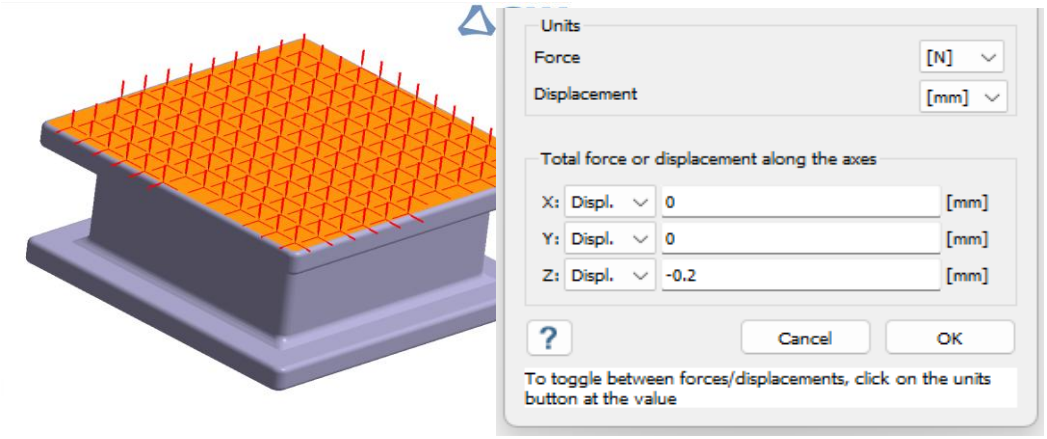
Geometry A- Block 20mm

Simsolid Settings		
		Units
Gap	0.096695	mm
Penetration	0.048348	mm
Connection resolution	Normal	-
Connect unwelded parts only	No	-
Regular connections	Bonded-top and bottom	-
Contact condition	Bonded	-
Constraint	Immovable-Bottom of bottom plate	
Refinement level	Steel-standard, Rubber-high	-
Max number of adaptive solutions	3	-
Adapt to thin solids	Yes	-
Adapt to features	No	-
Displacement and conditions	y=x=0 mm and z negative disp	

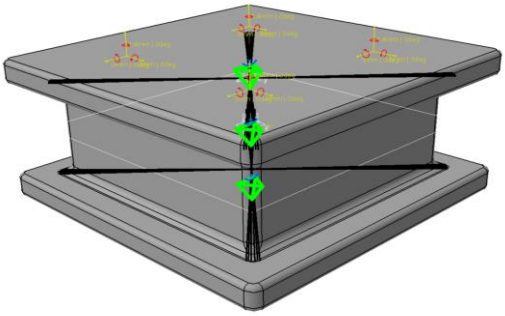
CATIA GAS Settings	
Analysis connection	General analysis connection both top and bottom
Properties	Rigid connection both top and bottom
Restraints	User-defined: All
	Clamp-bottom plates bottom
Loads	Enforced displacement-Translation 3
Sensor	Reaction sensor on the bottom rigid conn.
Force	Force 3

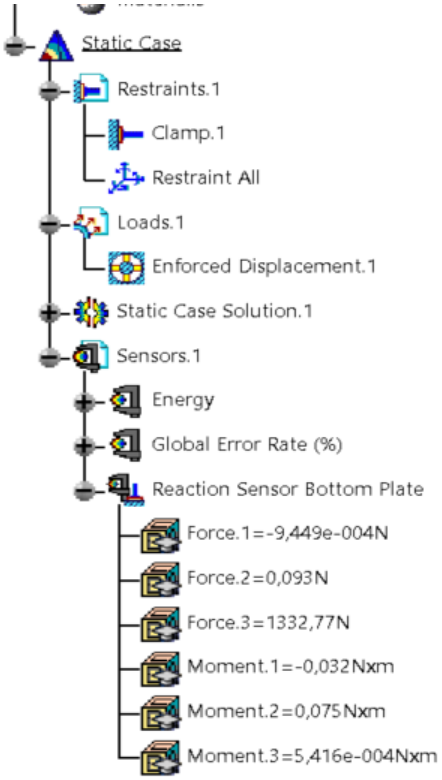
Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Simsolid-Rubber Special	F [N]	0	186.40	372.81	559.21	745.61	932.02	1118.4	1304.8
	d [mm]	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4



GAS-Rubber Special 5mm mesh, 0.5 sag	F [N]	0	190.396	380.791	571.187	761.583	951.979	1142.374	1332.77
	d [mm]	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4





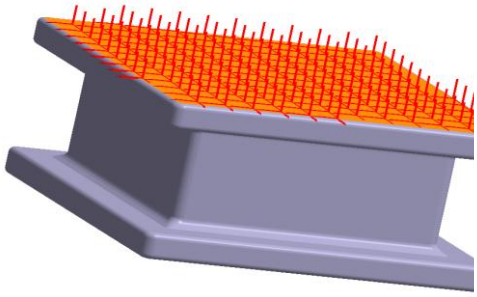
Geometry A- Block 20mm

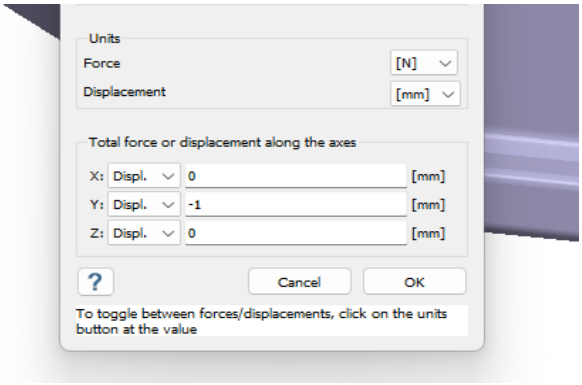
Simsolid Settings		
		Units
Gap	0.096695	mm
Penetration	0.048348	mm
Connection resolution	Normal	-
Connect unwelded parts only	No	-
Regular connections	Bonded-top and bottom	-
Contact condition	Bonded	-
Refinement level	Steel-standard, Rubber-high	-
Max number of adaptive solutions	3	-
Adapt to thin solids	Yes	-
Adapt to features	No	-
Displacement and conditions	z=x=0 mm and y negative disp	

CATIA GAS Settings	
Analysis connection	General analysis connection both top and bottom
Properties	Rigid connection both top and bottom
Restraints	User-defined: All
	Clamp-bottom plates bottom
Loads	Enforced displacement-Translation 2
Sensor	Reaction sensor on the bottom rigid conn.
Force	Force 2

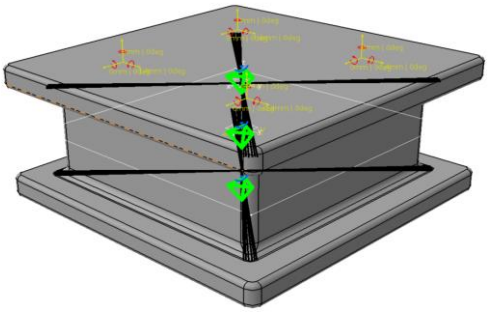
Youngs modulus	4.45	MPa
Poissons ratio	0.495	-

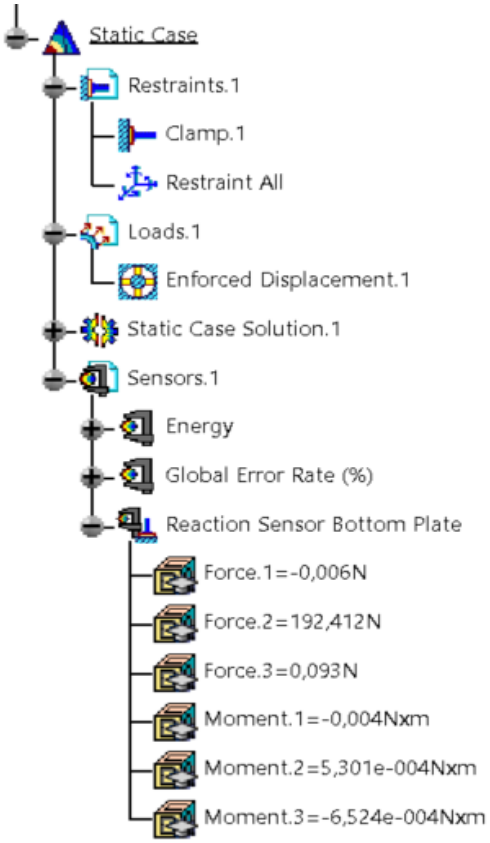
Simsolid-Rubber 60IR	F [N]	0	34.310	68.620	102.93	137.24	171.55	205.86	240.17
	d [mm]	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4





GAS-Rubber Special 5mm mesh, 0.5 sag	F [N]	0	34.378	68.757	103.135	137.513	171.892	206.27	240.648
	d [mm]	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4





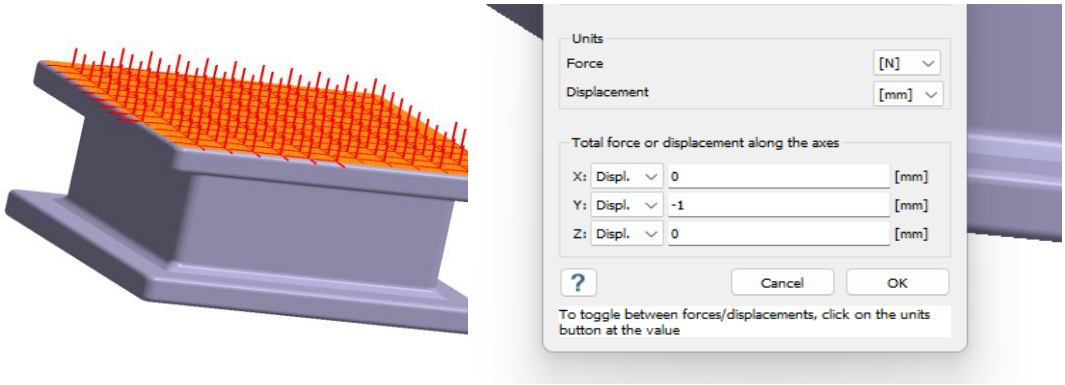
Geometry A- Block 20mm

Simsolid Settings		
		Units
Gap	0.096695	mm
Penetration	0.048348	mm
Connection resolution	Normal	-
Connect unwelded parts only	No	-
Regular connections	Bonded-top and bottom	-
Contact condition	Bonded	-
Refinement level	Steel-standard, Rubber-high	-
Max number of adaptive solutions	3	-
Adapt to thin solids	Yes	-
Adapt to features	No	-
Displacement and conditions	z=x=0 mm and y negative disp	

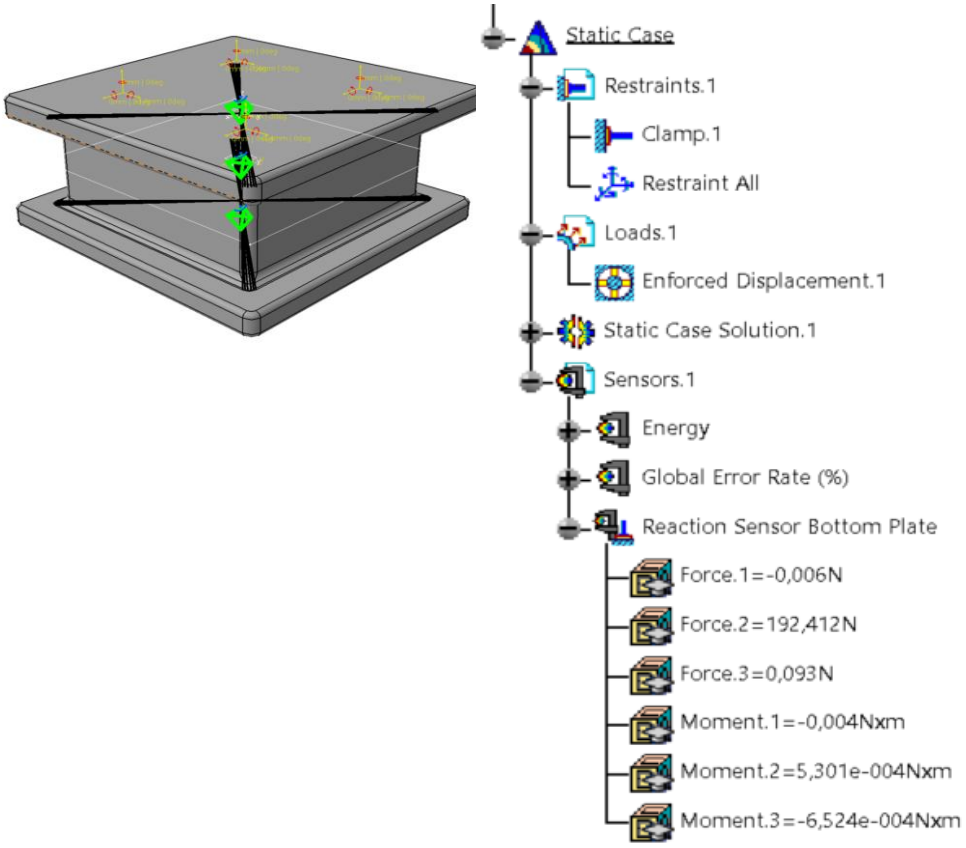
CATIA GAS Settings	
Analysis connection	General analysis connection both top and bottom
Properties	Rigid connection both top and bottom
Restraints	User-defined: All
	Clamp-bottom plates bottom
Loads	Enforced displacement-Translation 2
Sensor	Reaction sensor on the bottom rigid conn.
Force	Force 2

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Simsolid-Rubber 60IR	F [N]	0	27.307	54.615	81.922	109.23	136.54	163.84	191.15
	d [mm]	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4



GAS-Rubber Special 5mm mesh, 0.5 sag	F [N]	0	27.487	54.975	82.462	109.95	137.437	164.925	192.412
	d [mm]	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4



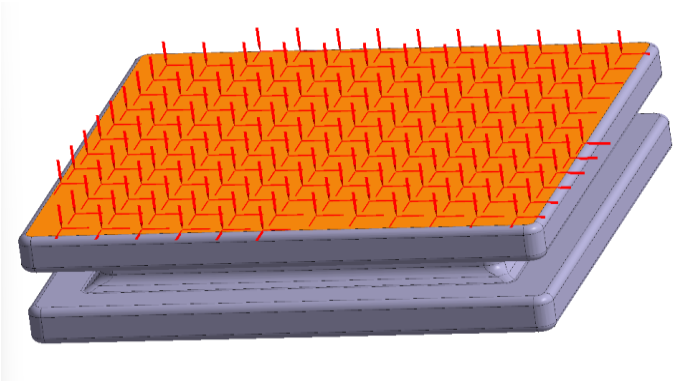
Geometry A- Block 5mm

Simsolid Settings		
		Units
Gap	0.09314	mm
Penetration	0.04657	mm
Connection resolution	Normal	-
Connect unwelded parts only	No	-
Regular connections	Bonded-top and bottom	-
Contact condition	Bonded	-
Constraint	Immovable-Bottom of bottom plate	
Refinement level	Steel-standard, Rubber-high	-
Max number of adaptive solutions	3	-
Adapt to thin solids	Yes	-
Adapt to features	No	-
Displacement and conditions	y=x=0 mm and z negative disp	

CATIA GAS Settings	
Analysis connection	General analysis connection both top and bottom
Properties	Rigid connection both top and bottom
Restraints	User-defined: All
	Clamp-bottom plates bottom
Loads	Enforced displacement-Translation 3
Sensor	Reaction sensor on the bottom rigid conn.
Force	Force 3

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

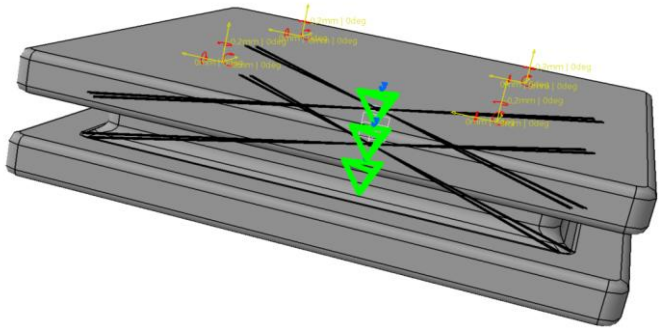
Simsolid-Rubber Special	F [N]	0	5273.0	####	15819	21092	26365
	d [mm]	0	0.2	0.4	0.6	0.8	1.0



GAS-Rubber Special	F [N]	0	6635.03	13270	19905.1(	26540..	33175.172
5mm mesh, 0.5 sag	d [mm]	0	0.2	0.4	0.6	0.8	1.0

GAS-Rubber Special	F [N]	0	5992.89	11985	17978.68	23971.!	29964.477
1mm mesh, 0.1 sag	d [mm]	0	0.2	0.4	0.6	0.8	1.0

GAS-Rubber Special	F [N]	0	5965.25	11930.513			
0.5mm mesh, 0.05 sag	d [mm]	0	0.2	0.4	0.6	0.8	1.0



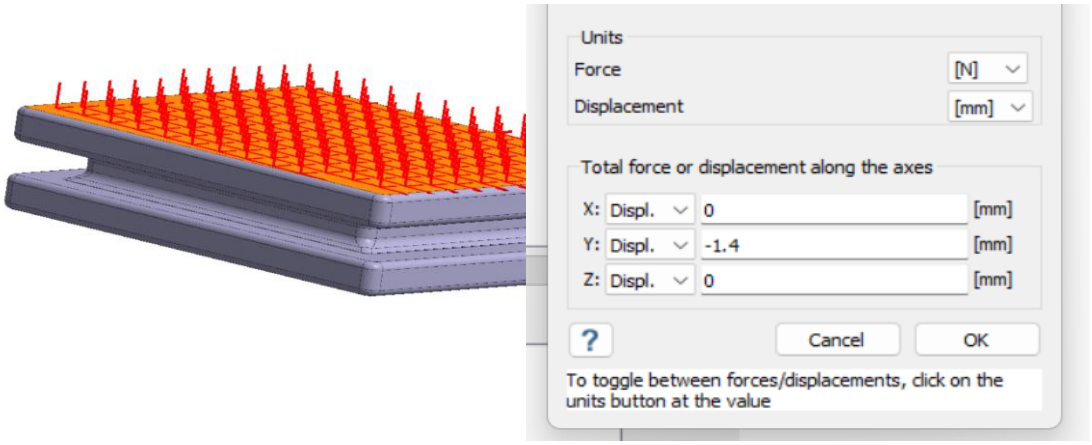
Geometry A- Block 5mm

Simsolid Settings		
		Units
Gap	0.09314	mm
Penetration	0.04657	mm
Connection resolution	Normal	-
Connect unwelded parts only	No	-
Regular connections	Bonded-top and bottom	-
Contact condition	Bonded	-
Refinement level	Steel-standard, Rubber-high	-
Max number of adaptive solutions	3	-
Adapt to thin solids	Yes	-
Adapt to features	No	-
Displacement and conditions	z=x=0 mm and y negative disp	

CATIA GAS Settings	
Analysis connection	General analysis connection both top and bottom
Properties	Rigid connection both top and bottom
Restraints	User-defined: All
	Clamp-bottom plates bottom
Loads	Enforced displacement-Translation 2
Sensor	Reaction sensor on the bottom rigid conn.
Force	Force 2

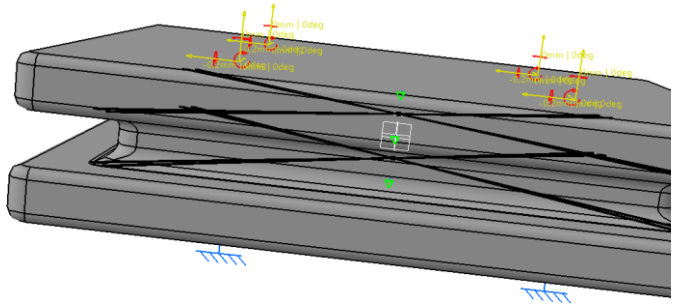
Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

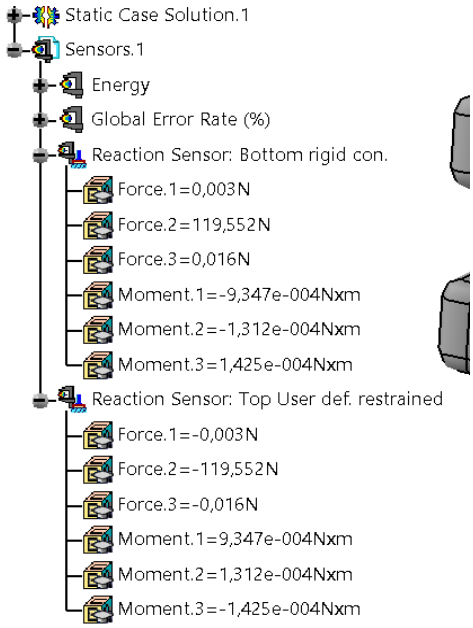
Simsolid-Rubber 60IR	F [N]	0	119.29	238.58	357.87	477.16	596.46	715.75	835.04
	d [mm]	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4



GAS-Rubber Special 5mm mesh, 0.5 sag	F [N]	0	120.299	240.598	360.897	481.196	601.495	721.794	842.093
	d [mm]	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4

GAS-Rubber Special 1mm mesh, 0.1 sag	F [N]	0	119.552	239.105	358.657	478.209	597.762	717.314	836.866
	d [mm]	0	0.2	0.4	0.6	0.8	1.0	1.2	1.4





Geometry A- Block 1mm

Simsolid Settings		
		Units
Gap		mm
Penetration		mm
Connection resolution	Normal	-
Connect unwelded parts only	No	-
Regular connections	Bonded-top and bottom	-
Contact condition	Bonded	-
Constraint	Immovable-Bottom of bottom plate	
Refinement level	Steel-standard, Rubber-high	-
Max number of adaptive solutions	3	-
Adapt to thin solids	Yes	-
Adapt to features	No	-
Displacement and conditions	y=x=0 mm and z negative disp	

CATIA GAS Settings	
Analysis connection	General analysis connection both top and bottom
Properties	Rigid connection both top and bottom
Restraints	User-defined: All
	Clamp-bottom plates bottom
Loads	Enforced displacement-Translation 3
Sensor	Reaction sensor on the bottom rigid conn.
Force	Force 3

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Simsolid-Rubber Special	F [N]	0	1189.7	2379.4	3569.2	4758.9	5948.6
	d [mm]	0	0.01	0.02	0.03	0.04	0.05

	11897	23794
0.1	0.2	

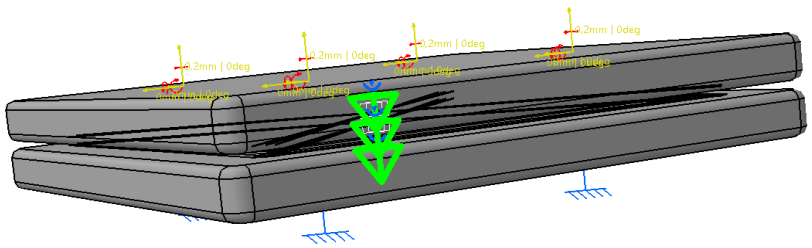
GAS-Rubber Special	F [N]	0	37970.11	75940.234	113910.352	151880.46	189850.58
5mm mesh, 0.5 sag	d [mm]	0	0.01	0.02	0.03	0.04	0.05

	379701	759402
0.1	0.2	

GAS-Rubber Special	F [N]	0	24852.82	49705.64	74558.46	99411.28	124264
1mm mesh, 0.1 sag	d [mm]	0	0.01	0.02	0.03	0.04	0.05

	248528	497056
0.1	0.2	

GAS-Rubber Special	F [N]	0	121948	
0.5mm mesh, 0.05 sag	d [mm]	0	0.05	0.1	0.2



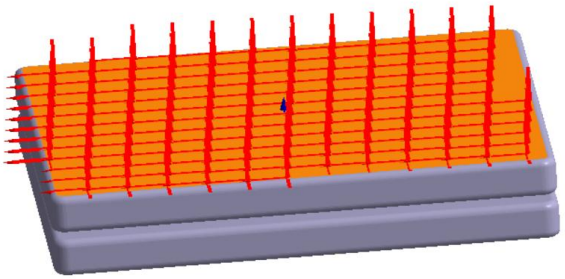
Geometry A- Block 1mm

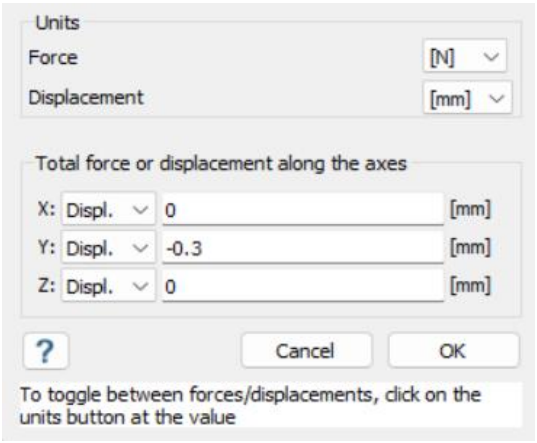
Simsolid Settings		
		Units
Gap	0.09258	mm
Penetration	0.04629	mm
Connection resolution	Normal	-
Connect unwelded parts only	No	-
Regular connections	Bonded-top and bottom	-
Contact condition	Bonded	-
Refinement level	Steel-standard, Rubber-high	-
Max number of adaptive solutions	3	-
Adapt to thin solids	Yes	-
Adapt to features	No	-
Displacement and conditions	z=x=0 mm and y negative disp	

CATIA GAS Settings	
Analysis connection	General analysis connection both top and bottom
Properties	Rigid connection both top and bottom
Restraints	User-defined: All
	Clamp-bottom plates bottom
Loads	Enforced displacement-Translation 2
Sensor	Reaction sensor on the bottom rigid conn.
Force	Force 2

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Simsolid-Rubber 60IR	F [N]	0	144.58	289.16	578.33	867.49
	d [mm]	0	0.05	0.1	0.2	0.3





GAS-Rubber Special 5mm mesh, 0.5 sag	F [N]	0	148.596	297.191	594.383	891.575
	d [mm]	0	0.05	0.1	0.2	0.3

GAS-Rubber Special 1mm mesh, 0.1 sag	F [N]	0	148.498	296.997	593.993	890.99
	d [mm]	0	0.05	0.1	0.2	0.3

Appendix B: Simulation results Geometry B

Geometry C- V-Style

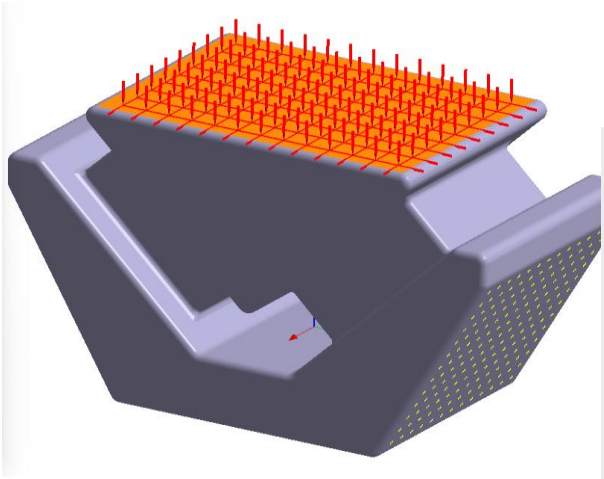
Simsolid Settings		
		Units
Gap	0.18451	mm
Penetration	0.092254	mm
Connection resolution	Normal	-
Connect unwelded parts only	No	-
Regular connections	Bonded-top and bottom both blocks	-
Contact condition	Bonded	-
Constraint	Immovable-Bottom of bottom plate and sides	
Refinement level	Steel-standard, Rubber-high	-
Max number of adaptive solutions	3	-
Adapt to thin solids	Yes	-
Adapt to features	No	-
Displacement and conditions	y=x=0 mm and z negative disp	

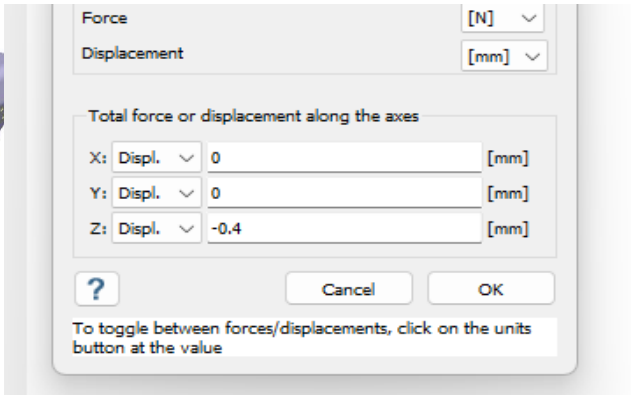
CATIA GAS Settings	
Analysis connection	General analysis connection both top and bottom
Properties	Rigid connection bottom
	Fastened connection top of the blocks
Restraints	User-defined: All
	Clamp-bottom sides
	Clamp-bottom plates bottom
Loads	Enforced displacement-Translation 3
Sensor	Reaction sensor on the bottom rigid conn.
Force	Force 3

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Simsolid-Rubber Special	F [N]	0	212.46	424.93	637.39	849.85	1062.3	1274.8
	d [mm]	0	0.2	0.4	0.6	0.8	1.0	1.2

1487.2	2124.6	3186.9	4249.3
1.4	2	3	4

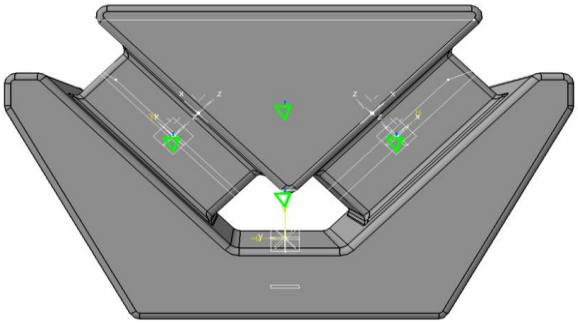




GAS-Rubber Special	F [N]	0	217.858	435.716	653.575	871.433	1089.291	1307.149
5mm mesh, 0.5 sag	d [mm]	0	0.2	0.4	0.6	0.8	1.0	1.2

The sensor is on the top, on the restraint

1525.00	2178.58	3267.873	4357.164
1.4	2	3	4



Appendix C: Simulation results Geometry C

Connections

GAS - disp z

Component list	
Plate Top	
Plate Bottom	
Sleeve	
Flange Bushing Top	
Flange Bushing Bottom	
Bracket	
Number of componentes	6
Number of connections	11

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Contact connection, friction 0.4
Flange Bushing Top-Bracket Horisontall	Rigid
Flaneg Bushing Top-Sleeve	Contact connection, friction 0.4
Flange Bushing Bottom-Bracket Verticall	Contact connection, friction 0.4
Flange Bushing Bottom-Bracket Horisontall	Rigid
Flange Bushing Bottom-Sleeve	Contact connection, friction 0.4
Flange Bushing Bottom- Plate Bottom	Rigid
Plate Bottom-Sleeve	Rigid
Flange Bushing Top- Bottom	Rigid

Youngs modulus	4.45	MPa			
Poissons ratio	0.495	-			
Full modell					
GAS-Rubber Special		F [N]	0	147.04	294.08
5mm mesh, 0.5 sag		d [mm]	0	0.02	0.04

Youngs modulus	3.5518	MPa			
Poissons ratio	0.4999	-			
Full modell					
GAS-Rubber Special		F [N]	0		259.437
5mm mesh, 0.5 sag		d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Contact connection, friction 0.4
Flange Bushing Top-Bracket Horisontall	Contact connection, friction 0.4
Flaneg Bushing Top-Sleeve	Contact connection, friction 0.4
Flange Bushing Bottom-Bracket Verticall	Contact connection, friction 0.4
Flange Bushing Bottom-Bracket Horisontall	Contact connection, friction 0.4
Flange Bushing Bottom-Sleeve	Contact connection, friction 0.4
Flange Bushing Bottom- Plate Bottom	Rigid
Plate Bottom-Sleeve	Rigid
Flange Bushing Top- Bottom	Rigid

GAS-Rubber Special	F [N]	0	
5mm mesh, 0.5 sag	d [mm]	0	0.02

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Contact connection, friction 0.4
Flange Bushing Top-Bracket Horisontall	Rigid
Flaneg Bushing Top-Sleeve	Contact connection, friction 0.4

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		136.638
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Contact connection, friction 0.4
Flange Bushing Top-Bracket Horisontall	Contact connection, friction 0.4
Flaneg Bushing Top-Sleeve	Contact connection, friction 0.4

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		104.889
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04

Relationship List	Connection type	Doesnt work
Plate top-Sleeve	Rigid	
Plate Top-Flange bushing Top	Contact connection, friction 0.4	
Flange Bushing Top-Bracket Verticall	Contact connection, friction 0.4	
Flange Bushing Top-Bracket Horisontall	Contact connection, friction 0.4	
Flaneg Bushing Top-Sleeve	Contact connection, friction 0.4	

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Slider connection
Flange Bushing Top-Bracket Verticall	Contact connection, friction 0.4
Flange Bushing Top-Bracket Horisontall	Contact connection, friction 0.4
Flaneg Bushing Top-Sleeve	Contact connection, friction 0.4

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Slider connection
Flange Bushing Top-Bracket Horisontall	Slider connection
Flaneg Bushing Top-Sleeve	Slider connection

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		72.007
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Fastened connection
Plate Top-Flange bushing Top	Fastened connection
Flange Bushing Top-Bracket Verticall	Slider connection
Flange Bushing Top-Bracket Horisontall	Slider connection
Flaneg Bushing Top-Sleeve	Slider connection

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		71.982
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Rigid
Flange Bushing Top-Bracket Horisontall	Rigid
Flaneg Bushing Top-Sleeve	Rigid

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		192.992
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Pressure fitting 0.05mm
Flange Bushing Top-Bracket Horisontall	Slider
Flaneg Bushing Top-Sleeve	Slider

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		144.817
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Pressure fitting 0.05mm
Flange Bushing Top-Bracket Horisontall	Contact connection, friction 0.4
Flaneg Bushing Top-Sleeve	Contact connection, friction 0.4

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		200.399
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Pressure fitting 0.1mm
Flange Bushing Top-Bracket Horisontall	Slider
Flaneg Bushing Top-Sleeve	Slider

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		194.575
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Smooth connection
Plate Top-Flange bushing Top	Smooth connection
Flange Bushing Top-Bracket Verticall	Pressure fitting 0.1mm
Flange Bushing Top-Bracket Horisontall	Slider
Flaneg Bushing Top-Sleeve	Slider

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		90.377
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Smooth connection
Plate Top-Flange bushing Top	Smooth connection
Flange Bushing Top-Bracket Verticall	Pressure fitting 0.05mm
Flange Bushing Top-Bracket Horisontall	Slider
Flaneg Bushing Top-Sleeve	Slider

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		67.967
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Smooth connection
Plate Top-Flange bushing Top	Smooth connection
Flange Bushing Top-Bracket Verticall	Pressure fitting 0.2mm
Flange Bushing Top-Bracket Horisontall	Slider
Flaneg Bushing Top-Sleeve	Slider

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		135.198
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Smooth connection
Plate Top-Flange bushing Top	Smooth connection
Flange Bushing Top-Bracket Verticall	Slider
Flange Bushing Top-Bracket Horisontall	Slider
Flaneg Bushing Top-Sleeve	Slider

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		38.238
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Slider
Flange Bushing Top-Bracket Horisontall	Slider
Flaneg Bushing Top-Sleeve	Rigid

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		72.535
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Slider
Flange Bushing Top-Bracket Horisontall	Rigid
Flaneg Bushing Top-Sleeve	Rigid

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		139.651
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Slider
Flange Bushing Top-Bracket Horisontall	Rigid
Flaneg Bushing Top-Sleeve	Slider

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		129.875
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Smooth Connection
Plate Top-Flange bushing Top	Smooth Connection
Flange Bushing Top-Bracket Verticall	Pressure fitting 0.1 mm
Flange Bushing Top-Bracket Horisontall	Pressure fitting 0.1 mm
Flaneg Bushing Top-Sleeve	Pressure fitting 0.1 mm

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		281.193
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Rigid
Flange Bushing Top-Bracket Horisontall	Rigid
Flaneg Bushing Top-Sleeve	Rigid

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		177.686
1mm mesh, 0.1 sag on rubber	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Slider
Flange Bushing Top-Bracket Horisontall	Slider
Flaneg Bushing Top-Sleeve	Slider

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		67.821
1mm mesh, 0.1 sag on rubber	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Contact connection fricition 0.4
Flange Bushing Top-Bracket Horisontall	Rigid
Flaneg Bushing Top-Sleeve	Rigid

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Cancelled after 9 hours

Half

GAS-Rubber Special	F [N]	0		
1mm mesh, 0.1 sag on rubber	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Sliding with friction 0.4
Flange Bushing Top-Bracket Verticall	Rigid
Flange Bushing Top-Bracket Horisontall	Rigid
Flaneg Bushing Top-Sleeve	Rigid

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

With bigger chafer or whats it called

Half

GAS-Rubber Special	F [N]	0		145.007
1mm mesh, 0.1 sag on rubber	d [mm]	0	0.02	0.04

Connections

GAS - disp y

Component list
Plate Top
Plate Bottom
Sleeve
Flange Bushing Top
Flange Bushing Bottom
Bracket

Number of componentes	6
Number of connections	11

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Rigid
Flange Bushing Top-Bracket Horisontall	Rigid
Flaneg Bushing Top-Sleeve	Rigid
Flange Bushing Bottom-Bracket Verticall	Rigid
Flange Bushing Bottom-Bracket Horisontall	Rigid
Flange Bushing Bottom-Sleeve	Rigid
Flange Bushing Bottom- Plate Bottom	Rigid
Plate Bottom-Sleeve	Rigid
Flange Bushing Top- Bottom	Rigid

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-
Full modell		

GAS-Rubber Special	F [N]	0		
1mm mesh, 0.1 sag on rubber	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Slider
Flange Bushing Top-Bracket Horisontall	Slider
Flaneg Bushing Top-Sleeve	Slider
Flange Bushing Bottom-Bracket Verticall	Slider
Flange Bushing Bottom-Bracket Horisontall	Slider
Flange Bushing Bottom-Sleeve	Slider
Flange Bushing Bottom- Plate Bottom	Slider
Plate Bottom-Sleeve	Rigid
Flange Bushing Top- Bottom	Rigid

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-
Full modell		

GAS-Rubber Special	F [N]	0		74.822
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Slider
Flange Bushing Top-Bracket Horisontall	Slider
Flaneg Bushing Top-Sleeve	Slider
Flange Bushing Bottom-Bracket Verticall	Slider
Flange Bushing Bottom-Bracket Horisontall	Slider
Flange Bushing Bottom-Sleeve	Slider
Flange Bushing Bottom- Plate Bottom	Slider
Plate Bottom-Sleeve	Rigid
Flange Bushing Top- Bottom	Rigid

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-
Full modell		

GAS-Rubber Special	F [N]	0		74.929
3mm mesh, 0.3 sag on rubber	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Pressure fitting 0.1mm
Flange Bushing Top-Bracket Horisontall	Slider
Flaneg Bushing Top-Sleeve	Slider
Flange Bushing Bottom-Bracket Verticall	Pressure fitting 0.1mm
Flange Bushing Bottom-Bracket Horisontall	Slider
Flange Bushing Bottom-Sleeve	Slider
Flange Bushing Bottom- Plate Bottom	Slider
Plate Bottom-Sleeve	Rigid
Flange Bushing Top- Bottom	Rigid

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-
Full modell		

GAS-Rubber Special	F [N]	0		
3mm mesh, 0.3 sag on rubber	d [mm]	0	0.02	0.04

Simplified Cases

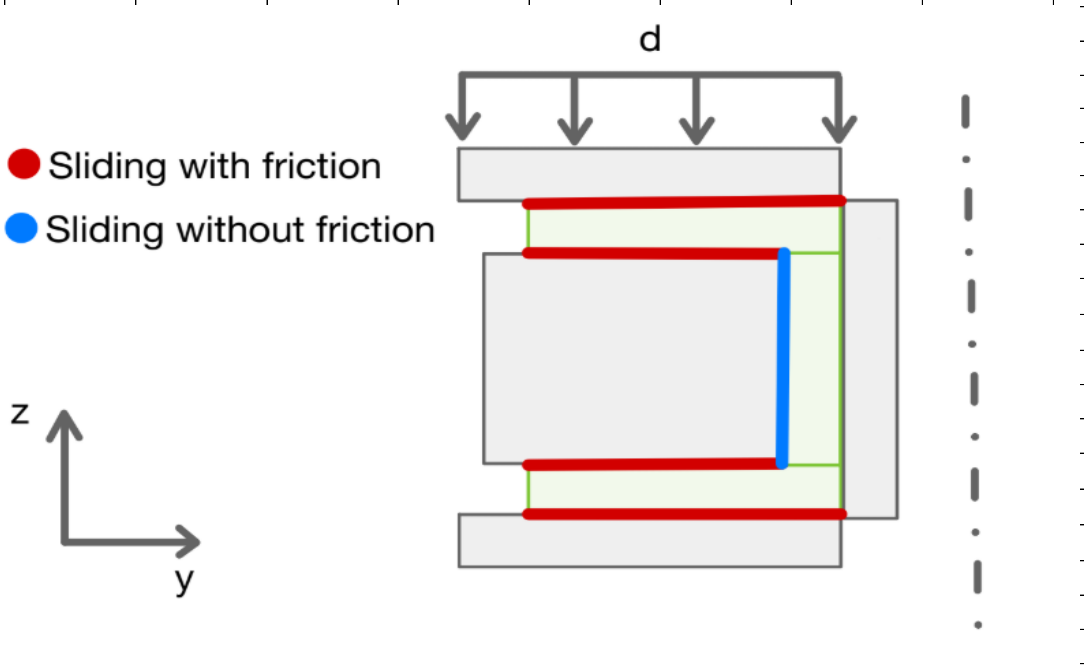
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Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Contact connection with friction 0.4
Flange Bushing Top-Bracket Verticall	
Flange Bushing Top-Bracket Horisontall	Contact connection with friction 0.4
Flaneg Bushing Top-Sleeve	Slider

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04

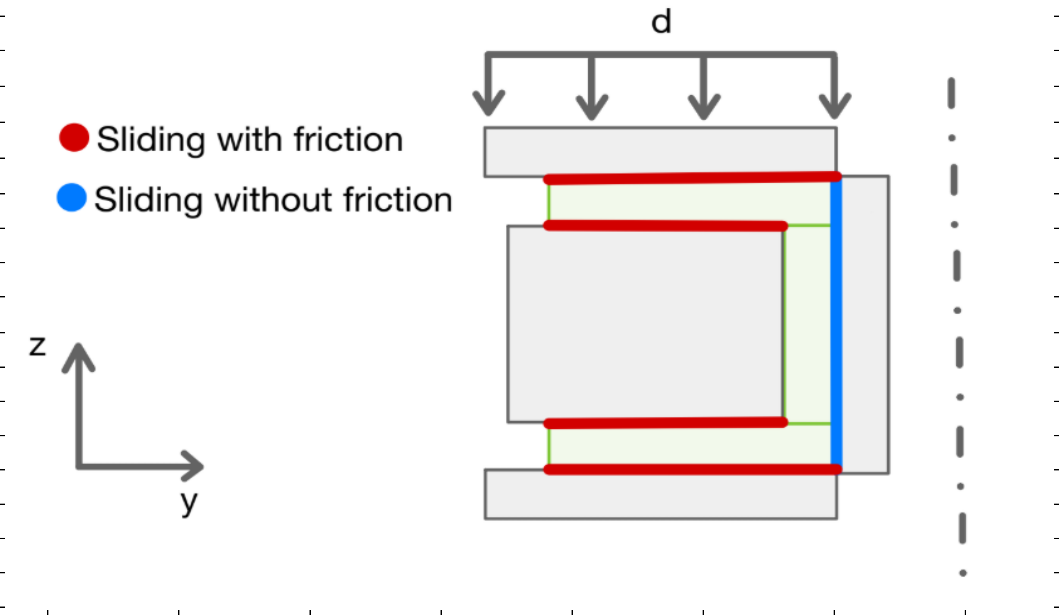


Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Contact connection with friction 0.4
Flange Bushing Top-Bracket Verticall	Slider
Flange Bushing Top-Bracket Horisontall	Contact connection with friction 0.4
Flaneg Bushing Top-Sleeve	

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		84.879
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04



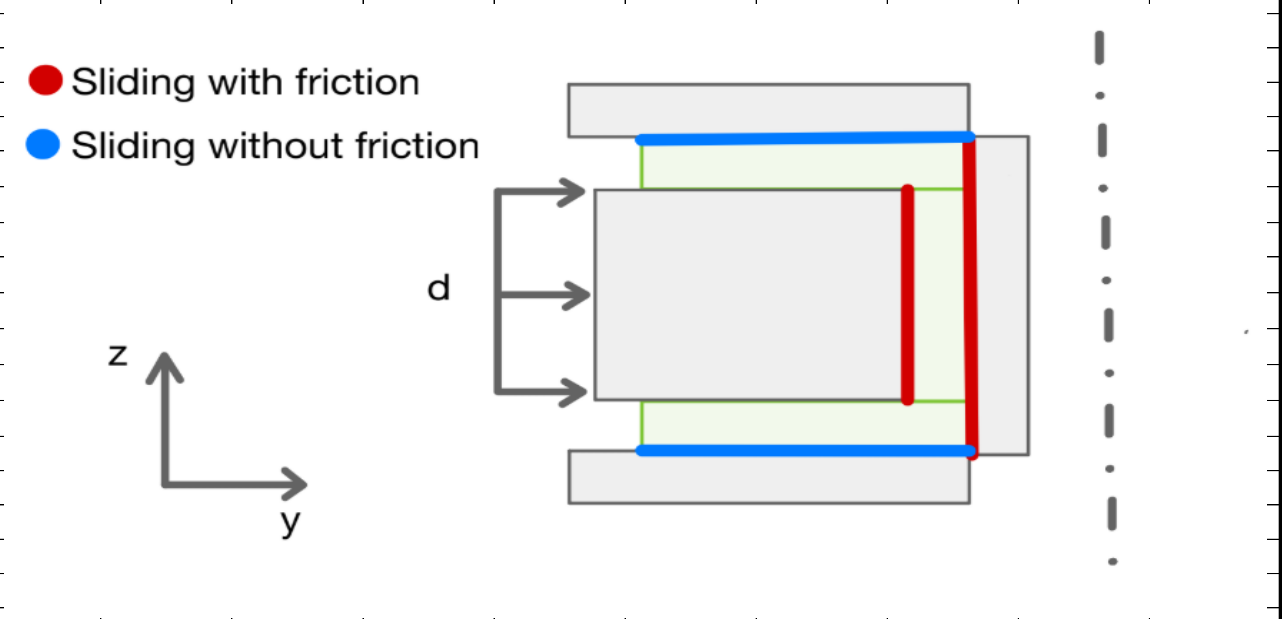
Disp y

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Sliding
Flange Bushing Top-Bracket Verticall	Contact connection with friction 0.4
Flange Bushing Top-Bracket Horisontall	
Flaneg Bushing Top-Sleeve	Contact connection with friction 0.4
Flange Bushing Bottom-Bracket Verticall	Contact connection with friction 0.4
Flange Bushing Bottom-Bracket Horisontall	
Flange Bushing Bottom-Sleeve	Contact connection with friction 0.4
Flange Bushing Bottom- Plate Bottom	Sliding
Plate Bottom-Sleeve	Rigid
Flange Bushing Top- Bottom	

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04

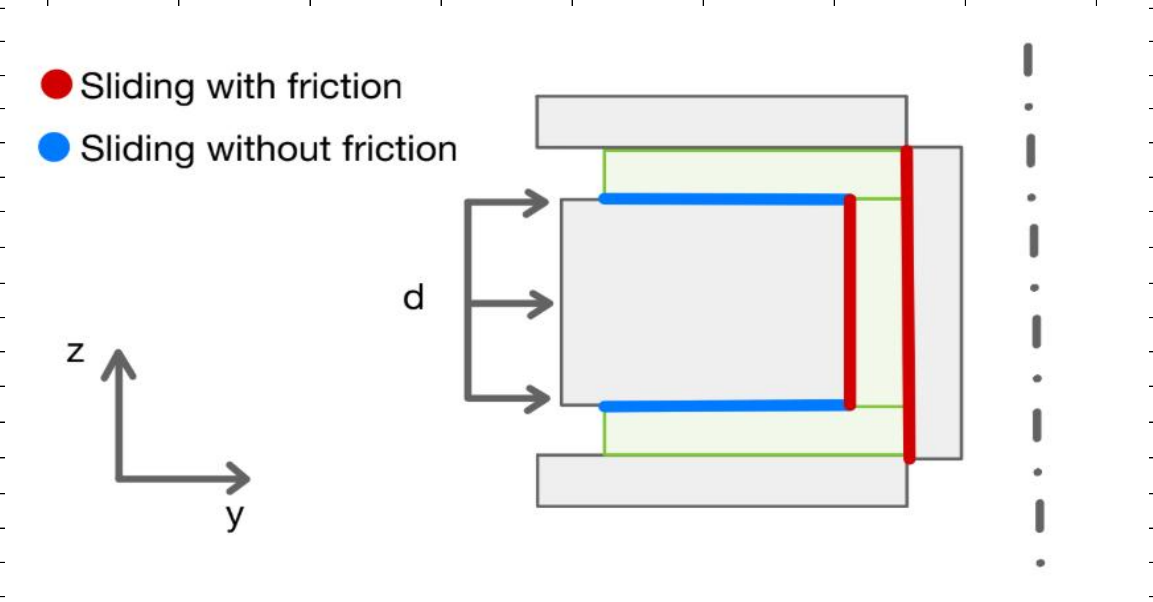


Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	
Flange Bushing Top-Bracket Verticall	Contact connection with friction 0.4
Flange Bushing Top-Bracket Horisontall	Sliding
Flaneg Bushing Top-Sleeve	Contact connection with friction 0.4
Flange Bushing Bottom-Bracket Verticall	Contact connection with friction 0.4
Flange Bushing Bottom-Bracket Horisontall	Sliding
Flange Bushing Bottom-Sleeve	Contact connection with friction 0.4
Flange Bushing Bottom- Plate Bottom	
Plate Bottom-Sleeve	Rigid
Flange Bushing Top- Bottom	

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

GAS-Rubber Special	F [N]	0		
5mm mesh, 0.5 sag	d [mm]	0	0.02	0.04



Disp z

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	
Flange Bushing Top-Bracket Horisontall	Rigid
Flaneg Bushing Top-Sleeve	

Youngs modulus	3.5518	MPa		Några sekunder	
Poissons ratio	0.4999	-			
Half					
GAS-Rubber Special		F [N]	0		2841.843
5mm mesh, 0.5 sag		d [mm]	0		1

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Sliding with friciton 0.4
Flange Bushing Top-Bracket Verticall	
Flange Bushing Top-Bracket Horisontall	Rigid
Flaneg Bushing Top-Sleeve	

Youngs modulus	3.5518	MPa			
Poissons ratio	0.4999	-			
Half					
GAS-Rubber Special		F [N]	0		2280.709
5mm mesh, 0.5 sag		d [mm]	0		1

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	
Flange Bushing Top-Bracket Horisontall	Sliding with friciton 0.4
Flaneg Bushing Top-Sleeve	

Youngs modulus	3.5518	MPa		40 minuter ish	
Poissons ratio	0.4999	-			
Half					
GAS-Rubber Special		F [N]	0		2300.867
5mm mesh, 0.5 sag		d [mm]	0		1

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Rigid
Flange Bushing Top-Bracket Horisontall	Rigid
Flaneg Bushing Top-Sleeve	Rigid
Flange Bushing Bottom-Bracket Verticall	Rigid
Flange Bushing Bottom-Bracket Horisontall	Rigid
Flange Bushing Bottom-Sleeve	Rigid
Flange Bushing Bottom- Plate Bottom	Rigid
Plate Bottom-Sleeve	Rigid
Flange Bushing Top- Bottom	

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

GAS-Rubber Special	F [N]	0		9084.75
5mm mesh, 0.5 sag	d [mm]	0		1

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Pressure fitting 0.5mm
Flange Bushing Top-Bracket Verticall	Rigid
Flange Bushing Top-Bracket Horisontall	Rigid
Flaneg Bushing Top-Sleeve	Pressure fitting 0.5mm
Flange Bushing Bottom-Bracket Verticall	Rigid
Flange Bushing Bottom-Bracket Horisontall	Rigid
Flange Bushing Bottom-Sleeve	Pressure fitting 0.5mm
Flange Bushing Bottom- Plate Bottom	Pressure fitting 0.5mm
Plate Bottom-Sleeve	Rigid
Flange Bushing Top- Bottom	Slider

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

GAS-Rubber Special	F [N]	0		7820.933
5mm mesh, 0.5 sag	d [mm]	0		1

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Pressure fitting 0.5mm
Flange Bushing Top-Bracket Verticall	Contact connection 0.4 friction
Flange Bushing Top-Bracket Horisontall	Contact connection 0.4 friction
Flaneg Bushing Top-Sleeve	Pressure fitting 0.5mm
Flange Bushing Bottom-Bracket Verticall	Contact connection 0.4 friction
Flange Bushing Bottom-Bracket Horisontall	Contact connection 0.4 friction
Flange Bushing Bottom-Sleeve	Pressure fitting 0.5mm
Flange Bushing Bottom- Plate Bottom	Pressure fitting 0.5mm
Plate Bottom-Sleeve	Rigid
Flange Bushing Top- Bottom	Slider

Youngs modulus	3.5518	MPa	failed three times
Poissons ratio	0.4999	-	

GAS-Rubber Special	F [N]	0		
5mm mesh, 0.5 sag	d [mm]	0		1

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Slider
Flange Bushing Top-Bracket Horisontall	Slider
Flaneg Bushing Top-Sleeve	Slider
Flange Bushing Bottom-Bracket Verticall	Slider
Flange Bushing Bottom-Bracket Horisontall	Slider
Flange Bushing Bottom-Sleeve	Rigid
Flange Bushing Bottom- Plate Bottom	Rigid
Plate Bottom-Sleeve	Slider
Flange Bushing Top- Bottom	

Youngs modulus	3.5518	MPa	
Poissons ratio	0.4999	-	

GAS-Rubber Special	F [N]	0	2463.614	3519.449
5mm mesh, 0.5 sag	d [mm]	0	0.7	1

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Slider
Flange Bushing Top-Bracket Verticall	Slider
Flange Bushing Top-Bracket Horisontall	Slider
Flaneg Bushing Top-Sleeve	Slider
Flange Bushing Bottom-Bracket Verticall	Slider
Flange Bushing Bottom-Bracket Horisontall	Slider
Flange Bushing Bottom-Sleeve	Rigid
Flange Bushing Bottom- Plate Bottom	Slider
Plate Bottom-Sleeve	Slider
Flange Bushing Top- Bottom	

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Singularity (didnt go th

GAS-Rubber Special	F [N]	0	
5mm mesh, 0.5 sag	d [mm]	0	1

Disp y

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Rigid
Flange Bushing Top-Bracket Horisontall	Rigid
Flaneg Bushing Top-Sleeve	Rigid
Flange Bushing Bottom-Bracket Verticall	Rigid
Flange Bushing Bottom-Bracket Horisontall	Rigid
Flange Bushing Bottom-Sleeve	Rigid
Flange Bushing Bottom- Plate Bottom	Rigid
Plate Bottom-Sleeve	Rigid
Flange Bushing Top- Bottom	

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

GAS-Rubber Special	F [N]	0		6449.462
5mm mesh, 0.5 sag	d [mm]	0		1

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Contact connection friction 0.4
Flange Bushing Top-Bracket Horisontall	Contact connection friction 0.4
Flaneg Bushing Top-Sleeve	Rigid
Flange Bushing Bottom-Bracket Verticall	Contact connection friction 0.4
Flange Bushing Bottom-Bracket Horisontall	Contact connection friction 0.4
Flange Bushing Bottom-Sleeve	Rigid
Flange Bushing Bottom- Plate Bottom	Rigid
Plate Bottom-Sleeve	Rigid
Flange Bushing Top- Bottom	

Youngs modulus	3.5518	MPa	Avbröt efter 7:45
Poissons ratio	0.4999	-	

GAS-Rubber Special	F [N]	0		
5mm mesh, 0.5 sag	d [mm]	0		1

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Rigid
Flange Bushing Top-Bracket Horisontall	Rigid
Flaneg Bushing Top-Sleeve	Rigid

Youngs modulus	3.5518	MPa	Några sekunder
Poissons ratio	0.4999	-	

Half

GAS-Rubber Special	F [N]	0		7749.602
5mm mesh, 0.5 sag	d [mm]	0		1

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	
Flange Bushing Top-Bracket Verticall	Contact connection friction 0.4
Flange Bushing Top-Bracket Horisontall	Contact connection friction 0.4
Flaneg Bushing Top-Sleeve	

Youngs modulus	3.5518	MPa	15 minuter kanske
Poissons ratio	0.4999	-	

Half

GAS-Rubber Special	F [N]	0		2.017e-7
5mm mesh, 0.5 sag	d [mm]	0		1

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	
Flange Bushing Top-Bracket Verticall	Rigid
Flange Bushing Top-Bracket Horisontall	
Flaneg Bushing Top-Sleeve	Rigid
Flange Bushing Bottom-Bracket Verticall	Rigid
Flange Bushing Bottom-Bracket Horisontall	
Flange Bushing Bottom-Sleeve	Rigid
Flange Bushing Bottom- Plate Bottom	
Plate Bottom-Sleeve	Rigid
Flange Bushing Top- Bottom	

Youngs modulus	3.5518	MPa	
Poissons ratio	0.4999	-	

GAS-Rubber Special	F [N]	0		4494.491
5mm mesh, 0.5 sag	d [mm]	0		1

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	
Flange Bushing Top-Bracket Verticall	Rigid
Flange Bushing Top-Bracket Horisontall	
Flaneg Bushing Top-Sleeve	Contact connection friction 0.4
Flange Bushing Bottom-Bracket Verticall	Rigid
Flange Bushing Bottom-Bracket Horisontall	
Flange Bushing Bottom-Sleeve	Contact connection friction 0.4
Flange Bushing Bottom- Plate Bottom	
Plate Bottom-Sleeve	Rigid
Flange Bushing Top- Bottom	

rough)

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

GAS-Rubber Special	F [N]	0		1369.289
5mm mesh, 0.5 sag	d [mm]	0		1

Med ändring på sleev och bracket med indent

Relationship List	Connection type
Plate top-Sleeve	Rigid
Plate Top-Flange bushing Top	Rigid
Flange Bushing Top-Bracket Verticall	Rigid
Flange Bushing Top-Bracket Horisontall	Rigid
Flaneg Bushing Top-Sleeve	Rigid
Flange Bushing Bottom-Bracket Verticall	Rigid
Flange Bushing Bottom-Bracket Horisontall	Rigid
Flange Bushing Bottom-Sleeve	Rigid
Flange Bushing Bottom- Plate Bottom	Rigid
Plate Bottom-Sleeve	Rigid
Flange Bushing Top- Bottom	Slider connection between

Youngs modulus	3.5518	MPa	Ökning av nästan dubbel beror på slider connection
Poissons ratio	0.4999	-	

GAS-Rubber Special	F [N]	0		12012.622
5mm mesh, 0.5 sag	d [mm]	0		1

Connections

Simsolid - disp z

Component list
Plate Top
Plate Bottom
Sleeve
Flange Bushing Top
Flange Bushing Bottom
Bracket

Number of componentes	6
Number of connections	11

Sleeve	Surpressed
Rubber Bottom	Surpressed
Bottom Plate	Surpressed

Relationship List	Connection type
Plate Top-Flange bushing Top	Sliding
Flange Bushing Top-Bracket Verticall	Sliding
Flange Bushing Top-Bracket Horisontall	Sliding

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Half

Simsolid Rubber	F [N]	0	51.452
	d [mm]	0 0.02	0.04

Relationship List	Connection type
Plate Top-Flange bushing Top	Seperating with friction 0.4
Flange Bushing Top-Bracket Verticall	Seperating with friction 0.4
Flange Bushing Top-Bracket Horisontall	Seperating with friction 0.4

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Also add to set up seperating contact

Half

Simsolid Rubber	F [N]	0	111.79
	d [mm]	0 0.02	0.04

Simplified Cases

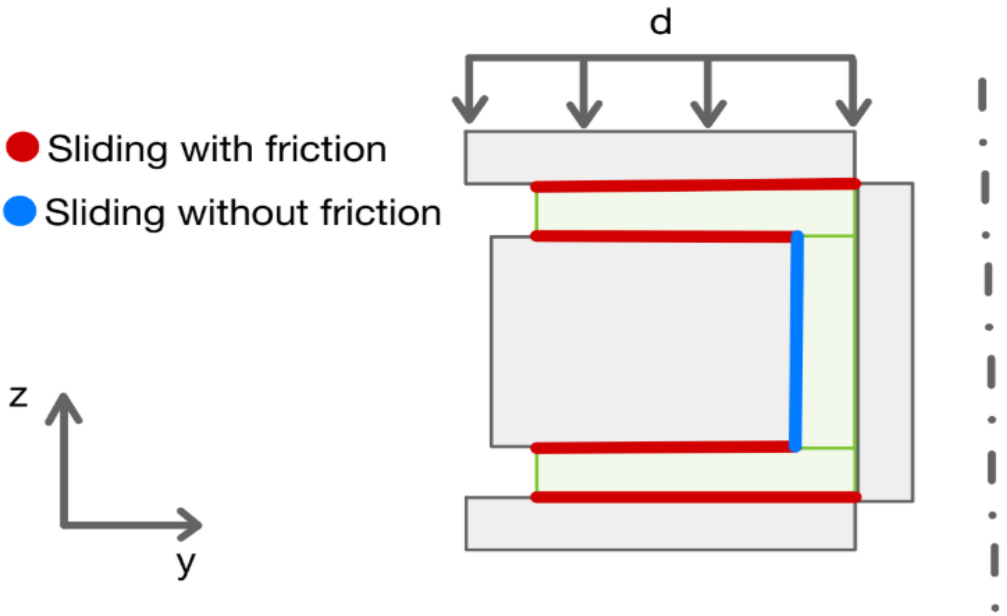
Disp z

Relationship List	Connection type
Plate top-Sleeve	Bonded
Plate Top-Flange bushing Top	Seperating with friction 0.4
Flange Bushing Top-Bracket Verticall	Sliding
Flange Bushing Top-Bracket Horisontall	Seperating with friction 0.4
Flaneg Bushing Top-Sleeve	
Flange Bushing Bottom-Bracket Verticall	Sliding
Flange Bushing Bottom-Bracket Horisontall	Seperating with friction 0.4
Flange Bushing Bottom-Sleeve	
Flange Bushing Bottom- Plate Bottom	Seperating with friction 0.4
Plate Bottom-Sleeve	Bonded
Flange Bushing Top- Bottom	

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

1.5 timmar ish

Simsolid Rubber	F [N]	0	2721.2
	d [mm]	0	1



Disp z

Relationship List	Connection type
Plate top-Sleeve	Bonded
Plate Top-Flange bushing Top	Seperating with friction 0.4
Flange Bushing Top-Bracket Verticall	Sliding
Flange Bushing Top-Bracket Horisontall	Seperating with friction 0.4
Flaneg Bushing Top-Sleeve	
Flange Bushing Bottom-Bracket Verticall	Sliding
Flange Bushing Bottom-Bracket Horisontall	Seperating with friction 0.4
Flange Bushing Bottom-Sleeve	
Flange Bushing Bottom- Plate Bottom	Seperating with friction 0.4
Plate Bottom-Sleeve	Bonded
Flange Bushing Top- Bottom	

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Simsolid Rubber	F [N]	0		
	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Bonded
Plate Top-Flange bushing Top	Bonded
Flange Bushing Top-Bracket Verticall	Bonded
Flange Bushing Top-Bracket Horisontall	Bonded
Flaneg Bushing Top-Sleeve	Bonded
Flange Bushing Bottom-Bracket Verticall	Bonded
Flange Bushing Bottom-Bracket Horisontall	Bonded
Flange Bushing Bottom-Sleeve	Bonded
Flange Bushing Bottom- Plate Bottom	Bonded
Plate Bottom-Sleeve	Bonded
Flange Bushing Top- Bottom	

Youngs modulus	3.5518	MPa	With inner sleeve
Poissons ratio	0.4999	-	

Simsolid Rubber	F [N]	0	9245.8
	d [mm]	0	1

Relationship List	Connection type
Plate top-Sleeve	Bonded
Plate Top-Flange bushing Top	Bonded
Flange Bushing Top-Bracket V	Sliding
Flange Bushing Top-Bracket H	Sliding
Flaneg Bushing Top-Sleeve	Sliding
Flange Bushing Bottom-Bracke	Sliding
Flange Bushing Bottom-Bracke	Sliding
Flange Bushing Bottom-Sleeve	Sliding
Flange Bushing Bottom- Plate I	Bonded
Plate Bottom-Sleeve	Bonded
Flange Bushing Top- Bottom	

Youngs modulus	3.551	MPa
Poissons ratio	0.49	-

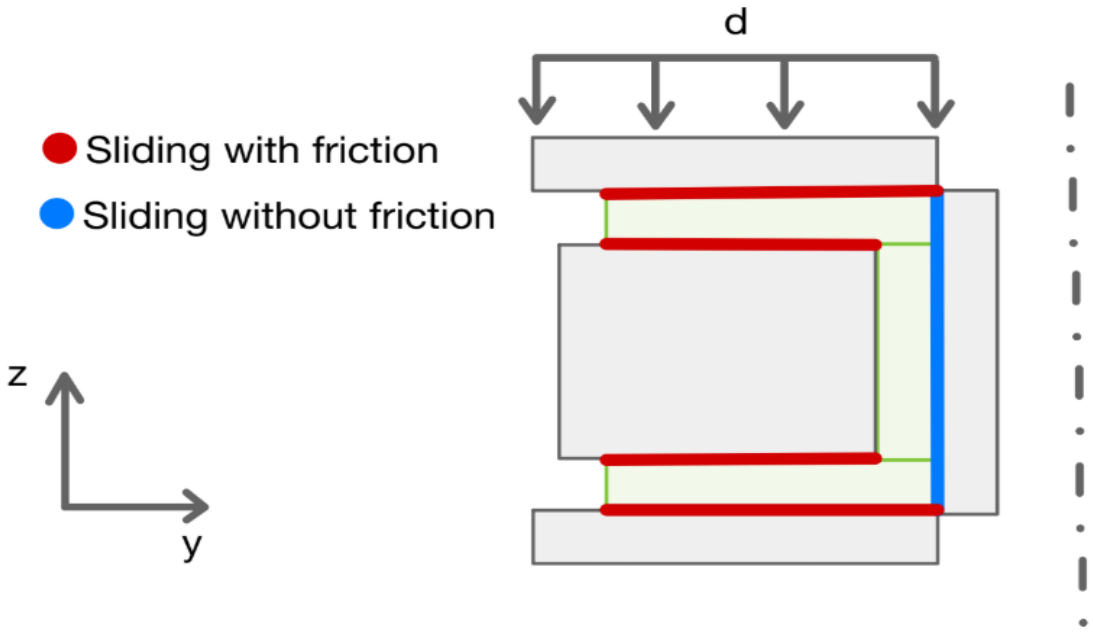
Full

Simsolid Rubber	F [N]	0	4582.9
	d [mm]	0	1

Relationship List	Connection type
Plate top-Sleeve	Bonded
Plate Top-Flange bushing Top	Sliding with friction 0.4
Flange Bushing Top-Bracket V	
Flange Bushing Top-Bracket H	Sliding with friction 0.4
Flaneg Bushing Top-Sleeve	Sliding
Flange Bushing Bottom-Bracke	
Flange Bushing Bottom-Bracke	Sliding with friction 0.4
Flange Bushing Bottom-Sleeve	Sliding
Flange Bushing Bottom- Plate I	Sliding with friction 0.4
Plate Bottom-Sleeve	Bonded
Flange Bushing Top- Bottom	

Youngs modulus	3.551	MPa
Poissons ratio	0.49	-

Simsolid Rubber	F [N]	0	3215.7
	d [mm]	0	1



Relationship List	Connection type
Plate top-Sleeve	Bonded
Plate Top-Flange bushing Top	Bonded
Flange Bushing Top-Bracket V	Bonded
Flange Bushing Top-Bracket H	Bonded
Flaneg Bushing Top-Sleeve	Bonded
Flange Bushing Bottom-Bracke	Bonded
Flange Bushing Bottom-Bracke	Bonded
Flange Bushing Bottom-Sleeve	Bonded
Flange Bushing Bottom- Plate I	Bonded
Plate Bottom-Sleeve	Bonded
Flange Bushing Top- Bottom	

Youngs modulus	3.551	MPa
Poissons ratio	0.49	-

Without inner sleeve

Simsolid Rubber	F [N]	0	3071.5
	d [mm]	0	1

Connections

Simsolid - disp y

Component list
Plate Top
Plate Bottom
Sleeve
Flange Bushing Top
Flange Bushing Bottom
Bracket

Number of componentes	6
Number of connections	11

Relationship List	Connection type
Plate top-Sleeve	Bonded
Plate Top-Flange bushing Top	Bonded
Flange Bushing Top-Bracket Verticall	Bonded
Flange Bushing Top-Bracket Horisontall	Bonded
Flaneg Bushing Top-Sleeve	Bonded
Flange Bushing Bottom-Bracket Verticall	Bonded
Flange Bushing Bottom-Bracket Horisontall	Bonded
Flange Bushing Bottom-Sleeve	Bonded
Flange Bushing Bottom- Plate Bottom	Bonded
Plate Bottom-Sleeve	Bonded
Flange Bushing Top- Bottom	Bonded

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Full

Simsolid Rubber	F [N]	0		376.45
	d [mm]	0	0.02	0.04

Relationship List	Connection type
Plate top-Sleeve	Bonded
Plate Top-Flange bushing Top	Bonded
Flange Bushing Top-Bracket Verticall	Sliding
Flange Bushing Top-Bracket Horisontall	Sliding
Flaneg Bushing Top-Sleeve	Sliding
Flange Bushing Bottom-Bracket Verticall	Sliding
Flange Bushing Bottom-Bracket Horisontall	Sliding
Flange Bushing Bottom-Sleeve	Sliding
Flange Bushing Bottom- Plate Bottom	Sliding
Plate Bottom-Sleeve	Bonded
Flange Bushing Top- Bottom	Bonded

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Full

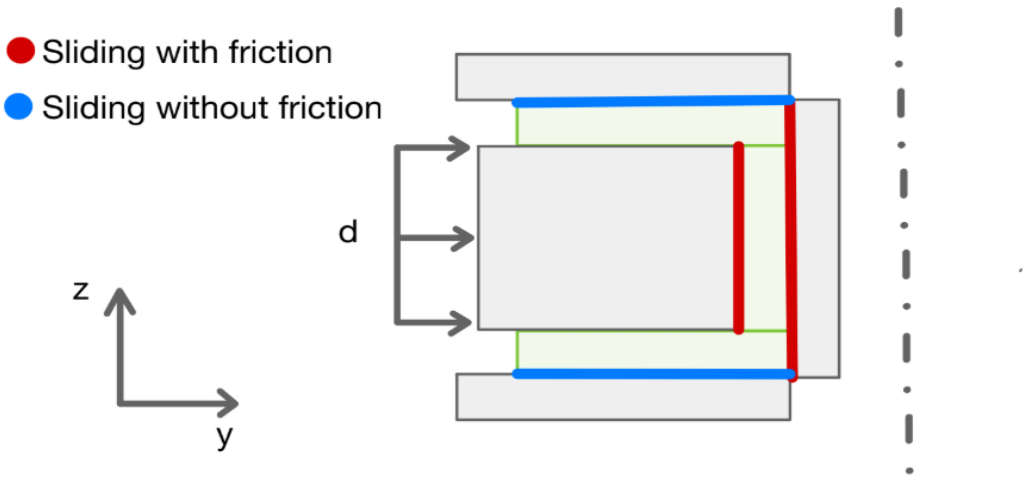
Simsolid Rubber	F [N]	0		152.92
	d [mm]	0	0.02	0.04

Disp y

Relationship List	Connection type
Plate top-Sleeve	Bonded
Plate Top-Flange bushing Top	Sliding
Flange Bushing Top-Bracket Verticall	Sliding with friction 0.4
Flange Bushing Top-Bracket Horisontall	
Flaneg Bushing Top-Sleeve	Sliding with friction 0.4
Flange Bushing Bottom-Bracket Verticall	Sliding with friction 0.4
Flange Bushing Bottom-Bracket Horisontall	
Flange Bushing Bottom-Sleeve	Sliding with friction 0.4
Flange Bushing Bottom- Plate Bottom	Sliding
Plate Bottom-Sleeve	Bonded
Flange Bushing Top- Bottom	

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Simsolid Rubber	F [N]	0	2305.9
	d [mm]	0	1



Disp y

Relationship List	Connection type
Plate top-Sleeve	Bonded
Plate Top-Flange bushing Top	Bonded
Flange Bushing Top-Bracket Verticall	Bonded
Flange Bushing Top-Bracket Horisontall	Bonded
Flaneg Bushing Top-Sleeve	Bonded
Flange Bushing Bottom-Bracket Verticall	Bonded
Flange Bushing Bottom-Bracket Horisontall	Bonded
Flange Bushing Bottom-Sleeve	Bonded
Flange Bushing Bottom- Plate Bottom	Bonded
Plate Bottom-Sleeve	Bonded
Flange Bushing Top- Bottom	

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

With inner sleeve

Simsolid Rubber	F [N]	0	9268.2
	d [mm]	0	1

Relationship List	Connection type
Plate top-Sleeve	Bonded
Plate Top-Flange bushing Top	Bonded
Flange Bushing Top-Bracket Verticall	Bonded
Flange Bushing Top-Bracket Horisontall	Bonded
Flaneg Bushing Top-Sleeve	Bonded
Flange Bushing Bottom-Bracket Verticall	Bonded
Flange Bushing Bottom-Bracket Horisontall	Bonded
Flange Bushing Bottom-Sleeve	Bonded
Flange Bushing Bottom- Plate Bottom	Bonded
Plate Bottom-Sleeve	Bonded
Flange Bushing Top- Bottom	

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Without inner sleeve

Simsolid Rubber	F [N]	0	748.38
	d [mm]	0	1

Relationship List	Connection type
Plate top-Sleeve	Bonded
Plate Top-Flange bushing Top	Bonded
Flange Bushing Top-Bracket Verticall	Sliding
Flange Bushing Top-Bracket Horisontall	Sliding
Flaneg Bushing Top-Sleeve	Sliding
Flange Bushing Bottom-Bracket Verticall	Sliding
Flange Bushing Bottom-Bracket Horisontall	Sliding
Flange Bushing Bottom-Sleeve	Sliding
Flange Bushing Bottom- Plate Bottom	Sliding
Plate Bottom-Sleeve	Bonded
Flange Bushing Top- Bottom	Bonded

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Full

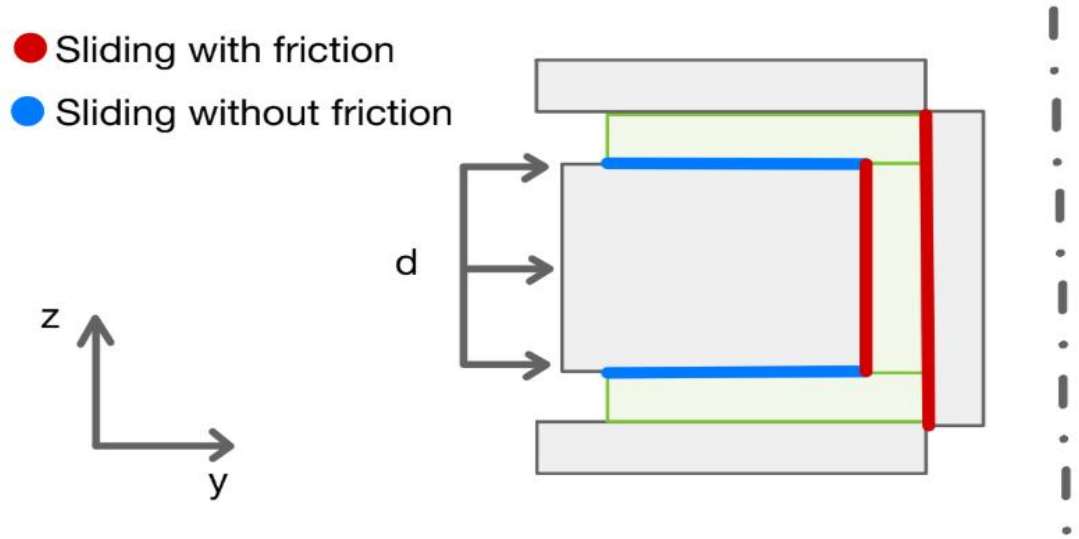
Simsolid Rubber	F [N]	0		71.837
	d [mm]	0	0.02	0.04

Sleeve	Surpressed

Relationship List	Connection type
Plate top-Sleeve	Bonded
Plate Top-Flange bushing Top	
Flange Bushing Top-Bracket Verticall	Sliding with friction 0.4
Flange Bushing Top-Bracket Horisontall	Sliding
Flaneg Bushing Top-Sleeve	Sliding with friction 0.4
Flange Bushing Bottom-Bracket Verticall	Sliding with friction 0.4
Flange Bushing Bottom-Bracket Horisontall	Sliding
Flange Bushing Bottom-Sleeve	Sliding with friction 0.4
Flange Bushing Bottom- Plate Bottom	
Plate Bottom-Sleeve	Bonded
Flange Bushing Top- Bottom	

Youngs modulus	3.5518	MPa
Poissons ratio	0.4999	-

Simsolid Rubber	F [N]	0	2516.6
	d [mm]	0	1



Appendix D: Poisson's ratio simulations

Youngs modulus	3.5518	MPa					
Poissons ratio	0.4999	-					
Simsolid-Rubber Special		F [N]	0			26365	
		d [mm]	0	0.6	0.8	1.0	
Youngs modulus	3.5518	MPa					
Poissons ratio	0.47	-					
Simsolid-Rubber Special		F [N]	0			7553.9	
		d [mm]	0	0.6	0.8	1.0	
Youngs modulus	3.5518	MPa					
Poissons ratio	0.45	-					
Simsolid-Rubber Special		F [N]	0			5440.6	
		d [mm]	0	0.6	0.8	1.0	
Youngs modulus	3.5518	MPa					
Poissons ratio	0.490	-					
Simsolid-Rubber Special		F [N]	0				13933
		d [mm]	0	0.2	0.6	0.8	1.0
Youngs modulus	3.5518	MPa					
Poissons ratio	0.4999	-					
GAS-Rubber Special		F [N]	0			29964.477	
1mm mesh, 0.1 sag		d [mm]	0	0.6	0.8	1.0	
Youngs modulus	3.5518	MPa					
Poissons ratio	0.490	-					
GAS-Rubber Special		F [N]	0			14376.222	
1mm mesh, 0.1 sag		d [mm]	0	0.6	0.8	1.0	
Youngs modulus	3.5518	MPa					
Poissons ratio	0.47	-					
GAS-Rubber Special		F [N]	0			7647.428	
1mm mesh, 0.1 sag		d [mm]	0	0.6	0.8	1.0	
Youngs modulus	3.5518	MPa					
Poissons ratio	0.45	-					
GAS-Rubber Special		F [N]	0			5477.491	
1mm mesh, 0.1 sag		d [mm]	0	0.6	0.8	1.0	